

**IT & CYBER
INTERNSHIP PROGRAM**

6-Week

Foundation + Specialization Learning Path

**20 Interns · 5 Specialization Tracks · 2-Week Common Foundation
NOC · SOC · Cloud (OpenStack) · Red Team / Pentesting · Cyber R&D**

Curriculum · Hands-on labs · Weekly KPIs · Capstone Projects

Prepared By Adil Shafiq — Manager Cloud & Security

July 2026

Table of Contents

Table of Contents.....	2
1. Program Overview	7
1.1 Program Structure at a Glance	7
1.2 The Two Phases	8
Phase 1 — Common Foundation (Weeks 1–2)	8
Phase 2 — Specialization (Weeks 3–6)	8
1.3 How to Read This Roadmap	8
2. Phase 1 — Common Foundation (Weeks 1–2)	9
2.1 Learning Objectives.....	9
2.2 Foundation Roadmap	9
2.3 Topic Breakdown	10
2.3.1 Operating Systems — Linux	10
2.3.2 Networking Fundamentals	10
2.3.3 Programming — Python	10
2.3.4 Version Control — Git & GitHub	11
2.3.5 Cybersecurity Fundamentals	11
2.3.6 Virtualization & Containers.....	12
2.3.7 Professional Skills.....	12
2.4 Week 1 Daily Schedule.....	13
2.5 Week 2 Daily Schedule.....	13
2.6 Weekly Assessment Checkpoints.....	13
2.7 Mini-Project — End of Week 2	14
"Secure Small Network + Automation Toolkit"	14
3. Phase 2 — Specialization Overview (Weeks 3–6)	15
3.1 How Track Selection Works	15
3.2 Track Comparison	15
4. Track 1 — NOC (Network Operations Center)	16
4.1 Learning Objectives.....	16
4.2 Required Tools	16
4.3 Track Roadmap	17
4.4 Topic Breakdown	18
4.4.1 Cisco & Enterprise Networking.....	18
4.4.2 Switching, VLANs & STP	18
4.4.3 Routing Protocols (OSPF)	18
4.4.4 ACLs & NAT	19
4.4.5 Firewalls & VPN.....	19

4.4.6 DNS / DHCP Services	19
4.4.7 Network Monitoring	19
4.4.8 Network Troubleshooting	20
4.5 Weekly Milestones	21
4.6 Sample Daily Activities	21
4.7 Mini-Projects	21
Mini-Project 1: Segmented, Access-Controlled Network (end of Week 4)	21
Mini-Project 2: Secure Branch-to-HQ Connectivity (end of Week 5)	21
4.8 Assessment Checkpoints	22
4.9 Capstone Project	22
Design, Deploy, Monitor & Troubleshoot an Enterprise Network	22
4.10 Career Outcomes	22
5. Track 2 — SOC (Security Operations Center)	23
5.1 Learning Objectives	23
5.2 Required Tools	23
5.3 Track Roadmap	24
5.4 Topic Breakdown	25
5.4.1 Blue Team & SOC Operations	25
5.4.2 Log Analysis & SIEM	25
5.4.3 MITRE ATT&CK & Threat Hunting	25
5.4.4 Sigma Rules & Detection Engineering	25
5.4.5 Incident Response	26
5.4.6 IOC Analysis & Malware Basics	26
5.4.7 Endpoint Monitoring	26
5.4.8 Threat Intelligence	27
5.5 Weekly Milestones	28
5.6 Sample Daily Activities	28
5.7 Mini-Projects	28
Mini-Project 1: Five Custom Detection Rules (end of Week 4)	28
Mini-Project 2: End-to-End Incident Report (end of Week 5)	28
5.8 Assessment Checkpoints	29
5.9 Capstone Project	29
Build a SOC Lab that Detects, Investigates & Reports Simulated Attacks	29
5.10 Career Outcomes	29
6. Track 3 — Cloud (OpenStack)	30
6.1 Learning Objectives	30
6.2 Required Tools	30
6.3 Track Roadmap	31

6.4 Topic Breakdown	32
6.4.1 Cloud Computing Fundamentals	32
6.4.2 OpenStack Architecture & Core Services	32
6.4.3 Deployment: DevStack & Kolla-Ansible	32
6.4.4 Networking, Security Groups & Floating IPs	32
6.4.5 Horizon, Heat & Orchestration	33
6.4.6 Advanced Services	33
6.4.7 High Availability & Multi-Node	33
6.4.8 Infrastructure as Code + Monitoring	34
6.5 Weekly Milestones.....	35
6.6 Sample Daily Activities	35
6.7 Mini-Projects.....	35
Mini-Project 1: Networked, Access-Controlled Tenant (end of Week 4)	35
Mini-Project 2: Heat-Templated Stack on a Multi-Node Cloud (end of Week 5)	35
6.8 Assessment Checkpoints.....	36
6.9 Capstone Project.....	36
Deploy & Secure a Production-Style Multi-Node OpenStack Cloud	36
6.10 Career Outcomes	36
7. Track 4 — Cybersecurity (Red Team / Pentesting)	37
7.1 Learning Objectives.....	37
7.2 Required Tools	37
7.3 Track Roadmap	38
7.4 Topic Breakdown	39
7.4.1 Reconnaissance & OSINT	39
7.4.2 Scanning & Enumeration (Nmap)	39
7.4.3 Web Exploitation (Burp Suite / OWASP Top 10).....	39
7.4.4 Active Directory Basics.....	39
7.4.5 Privilege Escalation (Linux & Windows).....	40
7.4.6 Password Attacks	40
7.4.7 Exploitation Frameworks	40
7.4.8 Reporting & Communication	41
7.5 Weekly Milestones.....	42
7.6 Sample Daily Activities	42
7.7 Mini-Projects.....	42
Mini-Project 1: Web Application Compromise Report (end of Week 4)	42
Mini-Project 2: Full Privilege-Escalation Chain — Linux & Windows (end of Week 5)	42
7.8 Assessment Checkpoints.....	43
7.9 Capstone Project.....	43

Full Penetration Test of a Lab Environment + Professional Report.....	43
7.10 Career Outcomes	43
8. Track 5 — R&D (Research & Development)	44
8.1 Learning Objectives.....	44
8.2 Required Tools	44
8.3 Track Roadmap	45
8.4 Topic Breakdown	46
8.4.1 Domain Orientation & Research Methodology	46
8.4.2 Domain: Artificial Intelligence & LLMs.....	46
8.4.3 Domain: Android Development	46
8.4.4 Domain: iOS Development	47
8.4.5 Domain: DevSecOps.....	47
8.4.6 Domain: Automation Engineering	47
8.5 Weekly Milestones.....	48
8.6 Sample Daily Activities	48
8.7 Mini-Projects.....	48
Mini-Project 1: Smallest Working Prototype (end of Week 4)	48
Mini-Project 2: Feature-Complete Prototype (end of Week 5)	48
8.8 Assessment Checkpoints.....	49
8.9 Capstone Project.....	49
Innovative Prototype — AI App, Mobile App, DevSecOps Pipeline, Automation Framework, or Research Demo	49
8.10 Career Outcomes	49
9. Intern KPI & Evaluation Templates	50
9.1 KPI Framework Overview.....	50
9.2 Weekly Intern KPI Scorecard (Universal — All Tracks)	51
9.3 Track-Specific Technical KPI Add-ons.....	52
Track 1 — NOC.....	52
Track 2 — SOC	52
Track 3 — Cloud (OpenStack)	52
Track 4 — Red Team	52
Track 5 — R&D.....	53
9.4 Mini-Project Evaluation Rubric (Template).....	54
9.5 Capstone Project Evaluation Rubric.....	55
9.6 Mid-Internship Review Form (End of Week 4)	56
9.7 Final Internship Performance Scorecard & Certificate Eligibility	57
Certificate Eligibility Threshold	57
9.8 Intern Self-Assessment Template (Weekly, Optional but Recommended)	58
9.9 Mentor / Supervisor Weekly Feedback Template	60

9.10 KPI Rating Scale & Grade Conversion	61
10. Appendices	62
Appendix A — Consolidated Master Resource List	62
Official Documentation.....	62
Free Courses & Learning Platforms	64
Hands-On Practice Platforms.....	64
Repositories	64
Certifications (Optional)	65
Appendix B — Required Tools & Software Checklist.....	66
Phase 1 — All Interns.....	66
Track 1 — NOC.....	66
Track 2 — SOC	66
Track 3 — Cloud (OpenStack)	66
Track 4 — Red Team.....	67
Track 5 — R&D.....	67
Appendix C — Glossary of Acronyms	68

1. Program Overview

This roadmap lays out a **6-week, two-phase learning path** for **interns** entering a IT & Cyber Internship Program. It is built in the visual style of colour-coded topic nodes, clear prerequisite flow, hands-on labs at every step, and curated external resources for deeper learning.

Field	Detail
Total Duration	6 weeks (2 weeks Common Foundation + 4 weeks Specialization)
Total Interns	20
Target Audience	Final-year university students or fresh graduates — beginner to intermediate technical level
Program Goal	Prepare interns for real-world industry roles through structured learning, extensive hands-on labs, and project-based learning
Learning Model	Phase 1: shared curriculum for all interns → Phase 2: 5 specialization tracks selected by fit & interest
Assessment Model	Daily labs, weekly quizzes, weekly KPI scorecards, mini-projects, and a final capstone per track (see Section 9)

1.1 Program Structure at a Glance

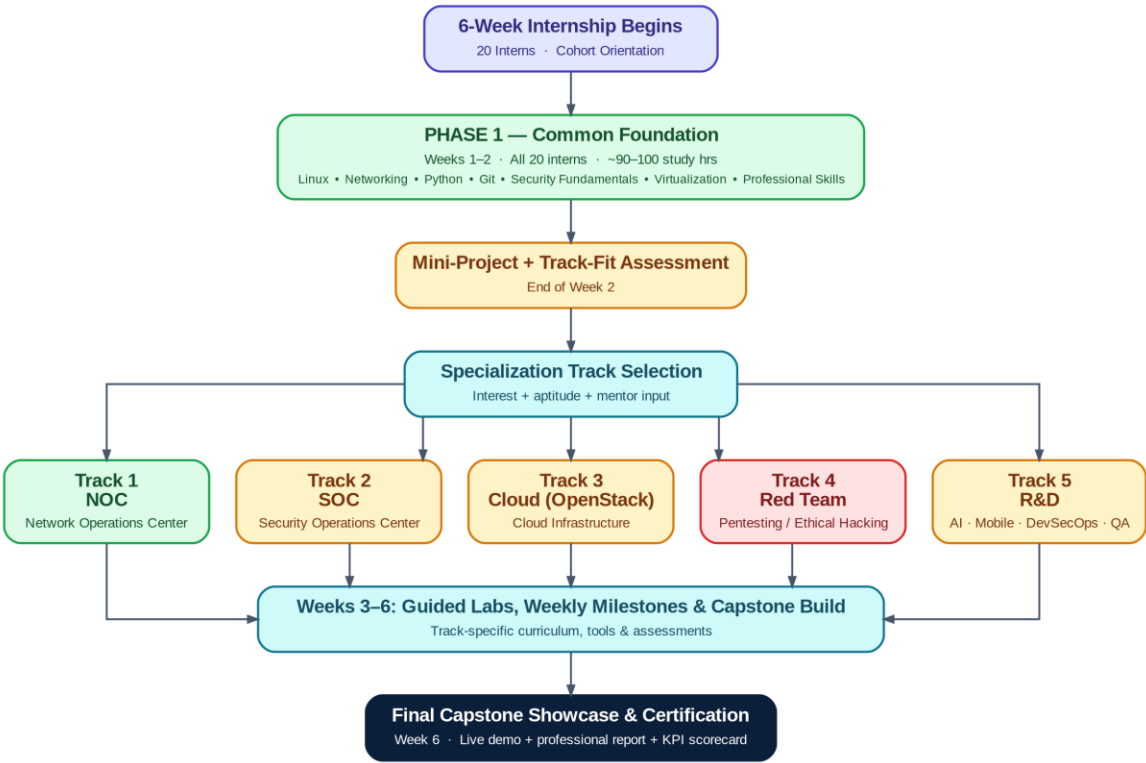


Figure 1 — End-to-end program flow: Common Foundation → Track Selection → Specialization → Capstone Showcase

1.2 The Two Phases

Phase 1 — Common Foundation (Weeks 1–2)

All 20 interns follow an identical curriculum covering Linux, networking, Python, Git, cybersecurity fundamentals, virtualization, and professional skills. This ensures every intern — regardless of specialization — shares a common technical baseline before splitting into tracks. Phase 1 ends with a graded mini-project and a track-fit conversation with mentors.

Phase 2 — Specialization (Weeks 3–6)

Interns split into five tracks aligned to real industry job families:

- **Track 1 — NOC** (Network Operations Center)
- **Track 2 — SOC** (Security Operations Center)
- **Track 3 — Cloud** (OpenStack-based private cloud engineering)
- **Track 4 — Cybersecurity** (Red Team / Penetration Testing)
- **Track 5 — R&D** (Research & Development — AI, Mobile, DevSecOps, or Automation)

Each track runs for 4 weeks with its own learning objectives, weekly milestones, hands-on labs, mini-projects, required tools, and a final capstone project mapped to specific job roles.

1.3 How to Read This Roadmap

Every topic node throughout this document follows the same format, so you can scan quickly: a coloured difficulty marker, a mandatory/optional badge, an estimated study-time budget, and (where relevant) a prerequisite. A worked example:

● INTERMEDIATE MANDATORY ~14h Prereq: Linux Fundamentals	
Marker	Meaning
● Beginner	No prior experience assumed beyond general computer literacy.
● Intermediate	Builds directly on Beginner topics earlier in the same track or Phase 1.
● Advanced	Requires comfort with the Intermediate material that precedes it; expect a steeper learning curve.
MANDATORY	Required for every intern in that track to reach capstone readiness.
OPTIONAL	Stretch content for interns who finish core material early or want deeper specialization.
~Nh	Estimated self-study + lab hours to reach working competency (excludes live instruction).
Prereq:	Topic(s) that should be completed first; "None" means it can start immediately.

Note on resources: to keep this roadmap usable rather than overwhelming, resources are grouped at the topic-cluster level rather than one link per micro-topic. Every cluster lists official documentation first, then free courses, video walkthroughs, hands-on practice platforms, repositories, and — where useful — books or certifications.

2. Phase 1 — Common Foundation (Weeks 1–2)

All 20 interns complete this identical two-week curriculum before selecting a specialization track. Topics build progressively — operating systems and networking first, then programming and version control, then security fundamentals and virtualization — so that every concept has the prerequisite knowledge it needs already in place.

2.1 Learning Objectives

- Operate confidently in a Linux command-line environment, including permissions, processes, and basic Bash automation.
- Explain and apply core networking concepts (OSI/TCP-IP, DNS/DHCP, routing/switching, VPNs) and use Wireshark to inspect real traffic.
- Write Python scripts that automate real tasks and interact with REST APIs.
- Use Git/GitHub professionally: branches, pull requests, and code review collaboration.
- Explain the CIA triad, authentication vs. authorization, encryption, hashing, and PKI — and apply basic vulnerability-management practice.
- Provision and use virtual machines and Docker containers as disposable lab environments.
- Produce clear technical documentation and work inside a lightweight Agile/Scrum cadence.

2.2 Foundation Roadmap

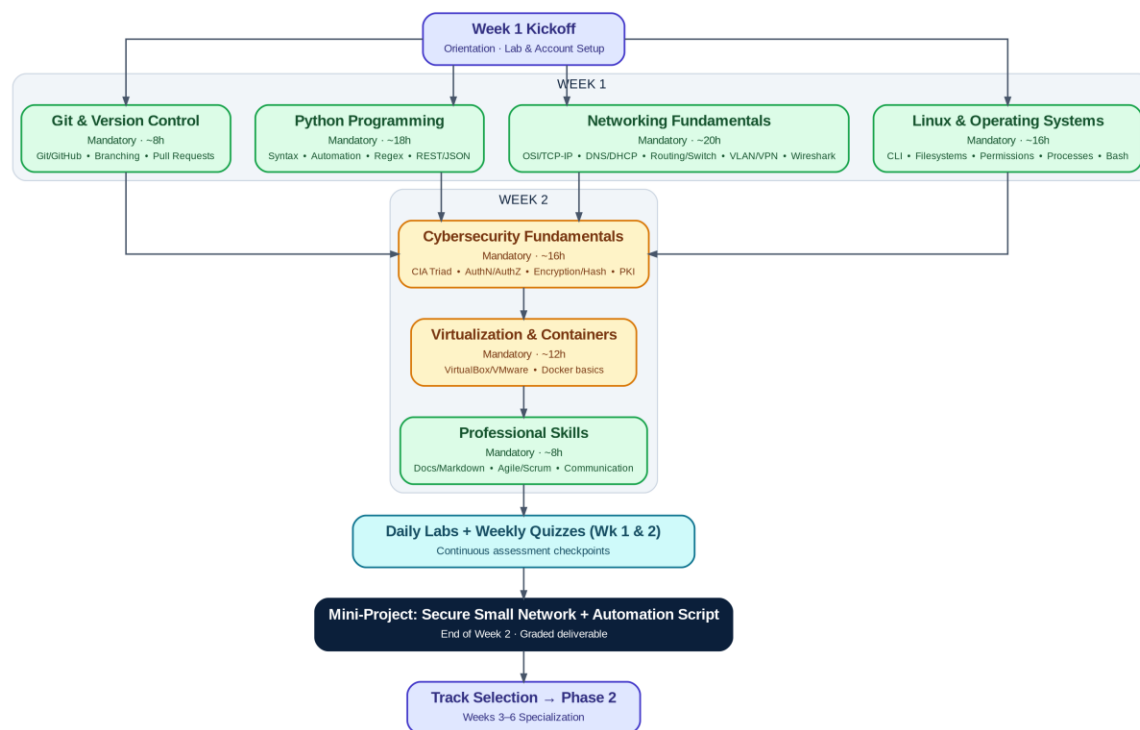


Figure 2 — Phase 1 topic flow across Week 1 and Week 2, converging on the Week 2 mini-project and track selection

2.3 Topic Breakdown

2.3.1 Operating Systems — Linux

● **BEGINNER** **MANDATORY** ~16h *Prereq: None*

The command line is the foundation every other Phase 1 topic (and every track) is built on.

Covers: Linux fundamentals • Linux command line • File systems • User & permission management • Process management • Bash scripting

Hands-on labs:

-
-
-

Resources:

- **Official Docs:** [Ubuntu Documentation](#)
- **Free Course:** [Linux Foundation — Introduction to Linux \(LFS101\)](#)
- **Video / YouTube:** [freeCodeCamp — Linux Crash Course \(with labs\)](#)
- **Hands-on / Practice:** [OverTheWire: Bandit wargame](#)
- **Book (optional):** [The Linux Command Line \(W. Shotts, free PDF\)](#)
- **Certification (optional):** [CompTIA Linux+](#)

2.3.2 Networking Fundamentals

● **BEGINNER** **MANDATORY** ~20h *Prereq: Linux (helps, not required)*

Covers the full stack from cabling to application-layer protocols, with Wireshark tying theory to real packets.

Covers: OSI Model • TCP/IP • IPv4 & IPv6 • DNS • DHCP • Routing • Switching • VLANs • VPNs • HTTP/HTTPS • SSH • SMTP • FTP • Firewalls • Wireshark

Hands-on labs:

-
-
-

Resources:

- **Official Docs:** [Cisco Networking Academy — Skills for All](#)
- **Free Course:** [freeCodeCamp — How Computer Networks Work \(9-hr, Network+ aligned\)](#)
- **Video / YouTube:** [freeCodeCamp.org YouTube channel](#)
- **Hands-on / Practice:** [Wireshark Sample Captures library](#)
- **Certification (optional):** [CompTIA Network+](#)

2.3.3 Programming — Python

● **BEGINNER** **MANDATORY** ~18h *Prereq: None*

Python is the automation language used across every track later in the program — this cluster gets everyone to a working baseline.

Covers: Python fundamentals • Python for automation • Regular expressions • APIs • JSON • REST

Hands-on labs:

-

-
-

Resources:

- **Official Docs:** [Python Official Documentation](#)
- **Free Course:** [freeCodeCamp — Python Full Course for Beginners](#)
- **Video / YouTube:** [freeCodeCamp — Networking in Python \(4 hands-on projects\)](#)
- **Hands-on / Practice:** [Real Python tutorials](#)
- **Certification (optional):** PCEP — Certified Entry-Level Python Programmer (Python Institute)

2.3.4 Version Control — Git & GitHub

● **BEGINNER** **MANDATORY** ~8h *Prereq: None*

Every deliverable from here forward — labs, mini-projects, capstones — is submitted through Git, so this cluster is scheduled early.

Covers: Git • GitHub • Branching • Pull Requests • Collaboration

Hands-on labs:

-
-

Resources:

- **Official Docs:** [Git Official Documentation](#)
- **Free Course:** [GitHub Docs — Get Started](#)
- **Video / YouTube:** freeCodeCamp.org YouTube — search "Git and GitHub for Beginners"
- **Hands-on / Practice:** [GitHub Skills \(interactive courses\)](#)
- **Repository:** [first-contributions \(beginner PR practice repo\)](#)

2.3.5 Cybersecurity Fundamentals

● **INTERMEDIATE** **MANDATORY** ~16h *Prereq: Networking Fundamentals*

The shared security vocabulary and mental model every track — NOC, SOC, Cloud, Red Team, R&D — builds on.

Covers: CIA Triad • Authentication • Authorization • Encryption • Hashing • PKI • Vulnerability Management • Security Best Practices

Hands-on labs:

-
-
-

Resources:

- **Official Docs:** [NIST Cybersecurity Framework \(CSF 2.0\)](#)
- **Free Course:** [Cisco — Introduction to Cybersecurity \(Skills for All\)](#)
- **Video / YouTube:** freeCodeCamp.org YouTube — search "Cyber Security Full Course"
- **Hands-on / Practice:** [TryHackMe — "Pre Security" learning path](#)
- **Certification (optional):** [CompTIA Security+](#)

2.3.6 Virtualization & Containers

● **INTERMEDIATE MANDATORY** ~12h *Prereq: Linux*

Every lab from Week 2 onward runs inside a VM or container, so interns need to be comfortable standing environments up and tearing them down.

Covers: VirtualBox • VMware • Docker fundamentals

Hands-on labs:

-
-

Resources:

- **Official Docs:** [Docker Official Documentation](#)
- **Free Course:** [Docker — Get Started Guide](#)
- **Video / YouTube:** freeCodeCamp.org YouTube — search "Docker Tutorial for Beginners"
- **Hands-on / Practice:** [Play with Docker \(browser-based sandbox\)](#)

2.3.7 Professional Skills

● **BEGINNER MANDATORY** ~8h *Prereq: None*

Technical skill without communication skill doesn't ship — this cluster is graded as part of every later deliverable, not just Phase 1.

Covers: Documentation • Technical writing • Markdown • Agile • Scrum • Communication • Presentation skills

Hands-on labs:

-
-

Resources:

- **Official Docs:** [Markdown Guide](#)
- **Free Course:** [Atlassian Agile Coach — Agile & Scrum 101](#)
- **Video / YouTube:** freeCodeCamp.org YouTube — search "Technical Writing"

2.4 Week 1 Daily Schedule

Day	Morning Focus	Afternoon Focus	Daily Lab / Deliverable
Mon	Orientation, account & lab-environment setup, cohort introductions	Linux basics: shell navigation, filesystem hierarchy	Document a Linux filesystem walkthrough (CLI-only)
Tue	Linux permissions & ownership	Process management + intro Bash scripting	Bash script: bulk-fix a broken permission tree
Wed	Networking: OSI model, TCP/IP, IPv4/IPv6	DNS, DHCP, routing & switching basics	Wireshark capture: identify 6+ protocols
Thu	VLANs, VPNs, firewalls, HTTP/HTTPS, SSH	Python fundamentals begin	Packet Tracer: build a mini routed/switched topology
Fri	Python for automation & regular expressions	Git & GitHub setup, first branch + pull request	Weekly Quiz 1 + automation script submitted via PR

2.5 Week 2 Daily Schedule

Day	Morning Focus	Afternoon Focus	Daily Lab / Deliverable
Mon	APIs, REST & JSON in Python	Cybersecurity fundamentals: CIA triad, authN/authZ	Script: consume a public REST API end-to-end
Tue	Encryption, hashing, PKI	Vulnerability management & security best practices	Hash-verification exercise + guided vulnerability scan
Wed	Virtualization: VirtualBox / VMware	Docker fundamentals	Provision 2 VMs + containerize a script
Thu	Professional skills: documentation, Markdown, Agile/Scrum	Mini-project build session (mentor office hours)	README + sprint board set up for mini-project
Fri	Mini-project completion & rehearsal	Mini-project demo day + track-fit conversations	Weekly Quiz 2 + Mini-Project live demo

2.6 Weekly Assessment Checkpoints

Checkpoint	Covers	Format	Weight toward Foundation KPI
Daily Labs (10 total)	That day's topic cluster	Graded pass/revise checklist, submitted via Git	40%
Weekly Quiz 1 (Fri, Wk 1)	Linux, Networking, Python basics, Git	20 MCQ/short-answer, 30 minutes	15%
Weekly Quiz 2 (Fri, Wk 2)	Security fundamentals, virtualization, professional skills	20 MCQ/short-answer, 30 minutes	15%
Mini-Project (end of Wk 2)	Integrates all 7 clusters	Working deliverable + live demo + README	30%

Full scoring rubrics and the printable Weekly KPI Scorecard used to record these checkpoints are provided in Section 9 — Intern KPI & Evaluation Templates.

2.7 Mini-Project — End of Week 2

"Secure Small Network + Automation Toolkit"

Working in pairs, interns design and document a small routed/switched lab network (Packet Tracer or GNS3), harden one Linux host on it (SSH hardening, firewall rules, least-privilege users), and write a Python automation script that checks the host's security posture (e.g., verifies firewall rules, flags weak permissions, or polls a REST API health check). The deliverable is submitted as a GitHub repository with a full README, and presented as a 5-minute live demo to mentors.

Required deliverables: (1) network topology diagram, (2) hardened VM with documented configuration, (3) working Python automation script with tests, (4) README.md, (5) 5-minute live demo.

Evaluation: scored against the Mini-Project Evaluation Rubric in Section 9.4, and feeds directly into each intern's track-fit conversation with mentors.

3. Phase 2 — Specialization Overview (Weeks 3–6)

At the start of Week 3, interns move from the shared Phase 1 curriculum into one of five specialization tracks. Track assignment is a three-way conversation between intern interest, demonstrated aptitude (from Phase 1 labs, quizzes, and the mini-project), and mentor guidance — not a purely first-come-first-served pick.

3.1 How Track Selection Works

- Interns rank their top 2 track preferences after the Week 2 mini-project demo.
- Mentors review each intern's Phase 1 KPI scorecard (Section 9.2) alongside stated preference.
- Where demand exceeds a track's cohort capacity, aptitude signals (relevant lab scores, quiz performance) break ties.
- Track 5 (R&D) additionally requires choosing a domain (or domains) within the first two days of Week 3 — see Section 8.
- Assignments are communicated by the Monday of Week 3 so every intern starts specialization content on time.

3.2 Track Comparison

Track	Focus	Est. Hours	Primary Job Roles
1 — NOC	Enterprise network design, operations & monitoring	~110–120h	NOC Technician/Analyst, Network Ops Engineer
2 — SOC	Detection, investigation & incident response	~110–120h	SOC Analyst (L1/L2), Threat Detection Analyst
3 — Cloud (OpenStack)	Private cloud deployment & automation	~115–125h	Cloud/OpenStack Engineer, Cloud Infrastructure Associate
4 — Red Team	Offensive security & penetration testing	~120–130h	Junior Penetration Tester, Red Team Associate
5 — R&D	AI, Mobile, DevSecOps, or Automation prototyping	~110–120h	Junior AI/ML Engineer, Mobile Developer, DevSecOps Engineer

Detailed learning objectives, weekly milestones, tools, resources, and capstones for each track follow in Sections 4–8.

4. Track 1 — NOC (Network Operations Center)

Keep enterprise networks designed, deployed, monitored, and fast to troubleshoot — the backbone role every other track depends on.

Field	Detail
Duration	4 weeks (Weeks 3–6)
Prerequisites	Phase 1 complete — especially Networking Fundamentals and Linux
Estimated Study Hours	~110–120 hours across 4 weeks
Target Job Roles	NOC Technician / Analyst · Network Operations Engineer · Junior Network Administrator · Network Support Engineer

4.1 Learning Objectives

- Configure and troubleshoot enterprise switching (VLANs, trunking, STP) and routing (static + OSPF) on real Cisco IOS devices or simulators.
- Implement access control (ACLs) and address translation (NAT/PAT) to secure and scale a network.
- Stand up a site-to-site VPN and a zone-based firewall policy.
- Deploy DNS/DHCP services and explain their role in enterprise operations.
- Build a monitoring stack (SNMP + Syslog + dashboards) that surfaces outages before users report them.
- Apply a structured troubleshooting methodology to diagnose and resolve multi-layer network faults.

4.2 Required Tools

Tool	Purpose
Cisco Packet Tracer	Primary simulator for device configuration and topology labs
GNS3	Advanced/near-production network emulation for capstone-level topologies
Wireshark	Packet capture and protocol-level troubleshooting
Zabbix	Infrastructure monitoring, alerting, and SLA dashboards
Grafana	Visualization layer for monitoring metrics
LibreNMS	Auto-discovery network monitoring (SNMP-based)

4.3 Track Roadmap

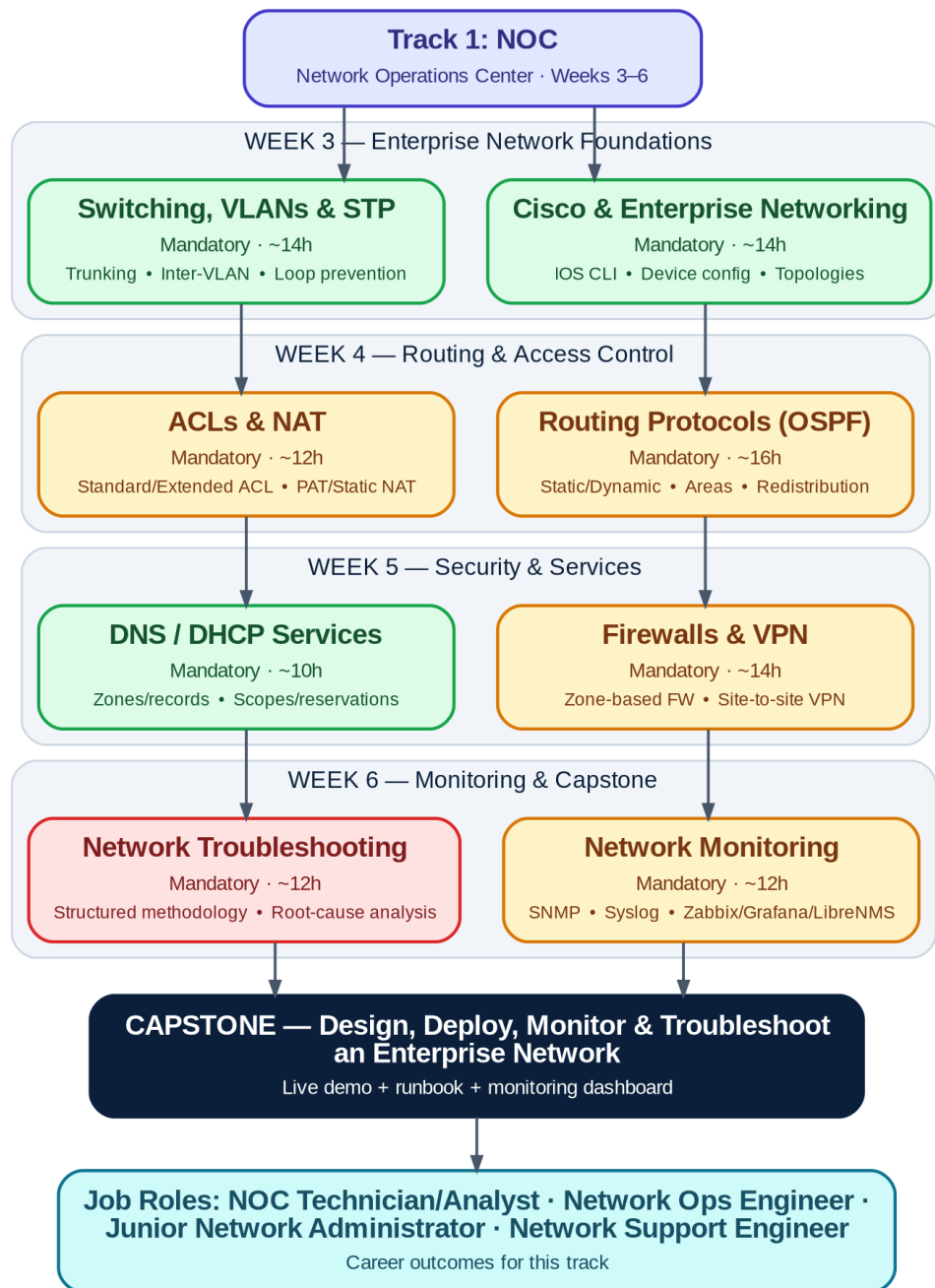


Figure 3 — NOC track roadmap: Weeks 3–6 progressing from Cisco fundamentals to enterprise capstone

4.4 Topic Breakdown

4.4.1 Cisco & Enterprise Networking

● **BEGINNER** **MANDATORY** ~14h *Prereq: Phase 1 Networking Fundamentals*

Translates OSI/TCP-IP theory into real Cisco IOS command-line practice on switches and routers.

Covers: Enterprise networking concepts • Cisco IOS CLI • Device configuration • Topology design

Hands-on labs:

-
-

Resources:

- **Official Docs:** [Cisco CCNA Certification Overview](#)
 - **Free Course:** [Cisco Networking Academy — Skills for All](#)
 - **Hands-on / Practice:** [GNS3 network simulator](#)
 - **Certification (optional):** [Cisco CCNA \(200-301\)](#)
-

4.4.2 Switching, VLANs & STP

● **BEGINNER** **MANDATORY** ~14h *Prereq: Cisco & Enterprise Networking*

Segmentation and loop-prevention — the two things that make switched networks scale safely.

Covers: Switching • VLANs • Trunking (802.1Q) • Inter-VLAN routing • Spanning Tree Protocol (STP)

Hands-on labs:

-
-

Resources:

- **Official Docs:** [Cisco Networking Academy — Learning Catalog](#)
 - **Free Course:** [CCNA: Introduction to Networks \(free course\)](#)
 - **Hands-on / Practice:** [Cisco Packet Tracer labs \(via Skills for All\)](#)
-

4.4.3 Routing Protocols (OSPF)

● **INTERMEDIATE** **MANDATORY** ~16h *Prereq: Switching, VLANs & STP*

Dynamic routing is what lets a network grow without every route being typed in by hand.

Covers: Static vs. dynamic routing • OSPF single-area • OSPF multi-area • Route redistribution basics

Hands-on labs:

-
-

Resources:

- **Official Docs:** [Cisco Networking Academy — Skills for All](#)
 - **Hands-on / Practice:** [GNS3 network simulator](#)
 - **Certification (optional):** [Cisco CCNA \(200-301\)](#)
-

4.4.4 ACLs & NAT

● **INTERMEDIATE MANDATORY** ~12h *Prereq: Routing Protocols (OSPF)*

The first line of traffic control on most enterprise edges — simple rules, high blast radius if misconfigured.

Covers: Standard ACLs • Extended ACLs • Static NAT • PAT (NAT overload)

Hands-on labs:

-
-

Resources:

- **Official Docs:** [Cisco Networking Academy — Skills for All](#)
 - **Hands-on / Practice:** [Cisco Packet Tracer labs](#)
-

4.4.5 Firewalls & VPN

● **INTERMEDIATE MANDATORY** ~14h *Prereq: ACLs & NAT*

Moves from basic packet filtering to policy-based security and secure site-to-site connectivity.

Covers: Zone-based firewall policy • Site-to-site VPN • IPsec basics

Hands-on labs:

-
-

Resources:

- **Official Docs:** [Cisco Networking Academy — Skills for All](#)
 - **Video / YouTube:** freeCodeCamp.org YouTube — search "VPN and firewall fundamentals"
-

4.4.6 DNS / DHCP Services

● **BEGINNER MANDATORY** ~10h *Prereq: Phase 1 Networking Fundamentals*

The two services that, when they fail, generate the most NOC tickets — worth mastering deeply.

Covers: DNS zones & records • DHCP scopes & reservations • Troubleshooting resolution failures

Hands-on labs:

-
-

Resources:

- **Official Docs:** [Cisco Networking Academy — Skills for All](#)
 - **Hands-on / Practice:** [GNS3 network simulator](#)
-

4.4.7 Network Monitoring

● **INTERMEDIATE MANDATORY** ~12h *Prereq: Firewalls & VPN, DNS/DHCP Services*

Turns a NOC from reactive to proactive — the difference between finding out from a user and finding out from a dashboard.

Covers: SNMP • Syslog • Zabbix • Grafana • LibreNMS

Hands-on labs:

-

-

Resources:

- **Official Docs:** [Zabbix Documentation](#)
 - **Free Course:** [Grafana Documentation](#)
 - **Hands-on / Practice:** [LibreNMS Documentation](#)
-

4.4.8 Network Troubleshooting

- **ADVANCED MANDATORY** ~12h *Prereq: Network Monitoring*

Integrates everything above into a repeatable diagnostic methodology instead of guesswork.

Covers: Structured troubleshooting methodology (OSI-layer approach) • Root-cause analysis • Documentation under pressure

Hands-on labs:

-
-

Resources:

- **Official Docs:** [Wireshark Documentation](#)
 - **Hands-on / Practice:** [Wireshark Sample Captures library](#)
-

4.5 Weekly Milestones

Week	Focus & Objective	Key Deliverable	Checkpoint
Wk 3	Cisco fundamentals, switching, VLANs & STP mastered on real device CLI	3-tier topology built + documented in Packet Tracer	Lab review + quiz
Wk 4	Dynamic routing (OSPF), ACLs, and NAT operational across multi-router topology	Mini-Project 1: Segmented, access-controlled network	Mini-project demo
Wk 5	Firewalls, VPN, and DNS/DHCP services deployed and hardened	Mini-Project 2: Secure branch-to-HQ connectivity	Mini-project demo + peer review
Wk 6	Monitoring stack live; troubleshooting methodology applied under injected-fault scenarios	Capstone: full enterprise network build	Capstone showcase

4.6 Sample Daily Activities

Representative week (Week 4) — the same daily rhythm (concept → guided lab → independent practice) repeats each week with new topics.

Day	Concept Block (AM)	Guided Lab (PM)	Independent Practice / Output
Mon	OSPF theory: areas, metrics, DR/BDR election	Configure single-area OSPF across 3 routers	Document routing table + explain convergence
Tue	OSPF troubleshooting patterns	Break a link; observe & explain reconvergence	Write a one-page incident note
Wed	ACL theory: standard vs. extended	Write & apply an extended ACL between VLANs	Test & document permitted/denied traffic
Thu	NAT/PAT theory	Configure PAT for outbound internet access	Verify translation table + peer walkthrough
Fri	Week 4 review + mentor office hours	Mini-Project 1 build & polish	Mini-Project 1 live demo + weekly quiz

4.7 Mini-Projects

Mini-Project 1: Segmented, Access-Controlled Network (end of Week 4)

Design a 3-VLAN network (Staff / Guest / Servers) with inter-VLAN routing, an extended ACL policy enforcing least-privilege access between segments, and PAT for shared internet egress.

Deliverable: Packet Tracer topology file + configuration exports + a one-page network diagram with ACL logic explained.

Mini-Project 2: Secure Branch-to-HQ Connectivity (end of Week 5)

Extend the Week 4 network with a simulated branch site connected via site-to-site VPN, protected by a zone-based firewall policy, with working DNS and DHCP services for both sites.

Deliverable: Updated topology + VPN/firewall configs + DNS/DHCP verification screenshots + short write-up.

4.8 Assessment Checkpoints

Checkpoint	Covers	Format	Weight
Weekly Labs (4 sets)	That week's cluster topics	Config export + verification screenshots, reviewed by mentor	35%
Weekly Quiz (×4)	Concepts from that week	20 MCQ/short-answer, 30 min	15%
Mini-Project 1 & 2	Weeks 4–5 integrated skills	Working topology + live walkthrough	25%
Capstone	Full track	Live demo + runbook + monitoring dashboard	25%

4.9 Capstone Project

Design, Deploy, Monitor & Troubleshoot an Enterprise Network

Acting as the NOC team for a fictional mid-size company opening a new branch office, interns (in pairs) design a 2-site enterprise network from scratch: HQ and branch, each multi-VLAN, connected via site-to-site VPN, protected by ACLs and a firewall policy, running DHCP/DNS, and fully monitored.

Requirements:

- Minimum 2 sites, 3+ VLANs per site, OSPF routing between all routed segments.
- Working site-to-site VPN and a documented firewall policy.
- DHCP and DNS services operational at both sites.
- Zabbix or LibreNMS + Grafana dashboard monitoring all devices with alert thresholds configured.
- A written runbook covering at least 3 common failure scenarios and their resolution steps.

Deliverables:

- Complete topology file (Packet Tracer or GNS3) with all configurations.
- Network diagram (logical + physical).
- Live monitoring dashboard.
- Runbook document.
- 15-minute live demo including one "mentor-injected fault" diagnosed live.

Evaluation: Scored against the Capstone Evaluation Rubric in Section 9.5 across design quality, security posture, monitoring completeness, troubleshooting performance, and documentation.

4.10 Career Outcomes

This track prepares interns for entry-level roles such as: **NOC Technician / Analyst** • **Network Operations Engineer** • **Junior Network Administrator** • **Network Support Engineer**

5. Track 2 — SOC (Security Operations Center)

Detect, investigate, and respond to attacks before they become breaches — the defensive front line of cybersecurity.

Field	Detail
Duration	4 weeks (Weeks 3–6)
Prerequisites	Phase 1 complete — especially Cybersecurity Fundamentals and Networking
Estimated Study Hours	~110–120 hours across 4 weeks
Target Job Roles	SOC Analyst (L1/L2) · Security Analyst · Threat Detection Analyst · Incident Response Associate

5.1 Learning Objectives

- Operate a SIEM (Wazuh / ELK / Splunk) to search, correlate, and triage security events.
- Map observed attacker behavior to MITRE ATT&CK tactics and techniques.
- Write and tune Sigma detection rules against real log sources.
- Run the incident-response lifecycle end to end: detect, contain, eradicate, recover, document.
- Triage indicators of compromise (IOCs) and perform first-pass malware analysis safely.
- Use endpoint telemetry (Sysmon, Velociraptor) and threat intelligence (VirusTotal) to enrich investigations.

5.2 Required Tools

Tool	Purpose
Wazuh	Open-source SIEM/XDR — log collection, detection rules, dashboards
ELK Stack	Log ingestion, search, and visualization (Elasticsearch, Logstash, Kibana)
Splunk	Enterprise SIEM — search processing language (SPL) practice
Microsoft Sentinel	Cloud-native SIEM/SOAR exposure
Velociraptor	Endpoint DFIR — forensic artifact collection at scale
VirusTotal	File/URL/IOC reputation and threat intelligence lookups
Sysmon	Windows endpoint telemetry for detection engineering

5.3 Track Roadmap

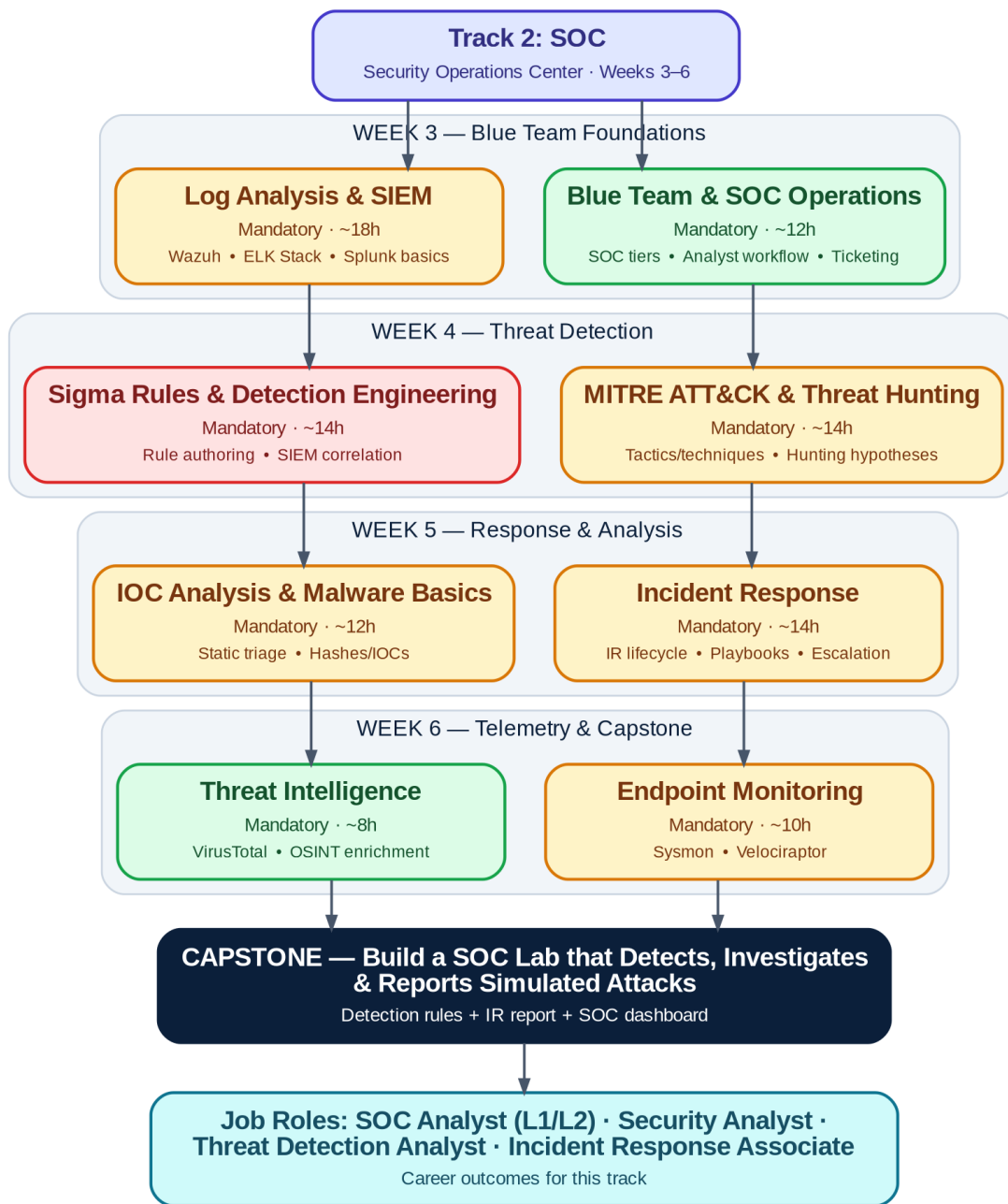


Figure 4 — SOC track roadmap: Weeks 3–6 progressing from Blue Team foundations to a live detection capstone

5.4 Topic Breakdown

5.4.1 Blue Team & SOC Operations

● **BEGINNER** **MANDATORY** ~12h *Prereq: Phase 1 Cybersecurity Fundamentals*

How a real SOC is organized — tiers, shift handoffs, ticketing — before touching any tooling.

Covers: SOC tiers (L1/L2/L3) • Analyst workflow • Ticketing & escalation

Hands-on labs:

-
-

Resources:

- **Official Docs:** [TryHackMe — SOC Level 1 learning path](#)
 - **Hands-on / Practice:** [LetsDefend — SOC analyst simulation platform](#)
-

5.4.2 Log Analysis & SIEM

● **INTERMEDIATE** **MANDATORY** ~18h *Prereq: Blue Team & SOC Operations*

The daily-driver skill of a SOC analyst: pulling signal out of an overwhelming volume of logs.

Covers: Wazuh • ELK Stack (Elasticsearch/Logstash/Kibana) • Splunk SPL basics

Hands-on labs:

-
-

Resources:

- **Official Docs:** [Wazuh Documentation](#)
 - **Free Course:** [Elastic \(ELK\) Documentation](#)
 - **Hands-on / Practice:** [Splunk Documentation](#)
-

5.4.3 MITRE ATT&CK & Threat Hunting

● **INTERMEDIATE** **MANDATORY** ~14h *Prereq: Log Analysis & SIEM*

Gives analysts a shared vocabulary for adversary behavior and a starting point for proactive hunts.

Covers: Tactics & techniques • Sub-techniques • Building a hunting hypothesis

Hands-on labs:

-
-

Resources:

- **Official Docs:** [MITRE ATT&CK — Enterprise Matrix](#)
 - **Hands-on / Practice:** [MITRE ATT&CK Navigator](#)
-

5.4.4 Sigma Rules & Detection Engineering

● **ADVANCED** **MANDATORY** ~14h *Prereq: MITRE ATT&CK & Threat Hunting*

Moves interns from consuming detections to writing and tuning their own — the SOC L2 skill.

Covers: Sigma rule syntax • Rule authoring • SIEM correlation & tuning false positives

Hands-on labs:

-
-

Resources:

- **Official Docs:** [SigmaHQ — Main Rule Repository](#)
 - **Free Course:** [Sigma — Resources & Getting Started](#)
-

5.4.5 Incident Response

● **INTERMEDIATE** **MANDATORY** ~14h *Prereq: Sigma Rules & Detection Engineering*

The structured process that turns a detected alert into a closed, documented incident.

Covers: IR lifecycle (prepare, detect, contain, eradicate, recover, learn) • Playbooks • Escalation criteria

Hands-on labs:

-
-

Resources:

- **Official Docs:** [NIST — Incident Handling Guidance \(SP 800-61\)](#)
 - **Hands-on / Practice:** [TryHackMe — SOC Level 1 learning path](#)
-

5.4.6 IOC Analysis & Malware Basics

● **INTERMEDIATE** **MANDATORY** ~12h *Prereq: Incident Response*

First-pass triage skills for suspicious files and indicators — enough to decide escalate vs. close.

Covers: Static triage • File hashes & IOCs • Safe-handling basics

Hands-on labs:

-
-

Resources:

- **Official Docs:** [VirusTotal](#)
 - **Hands-on / Practice:** [VirusTotal](#)
-

5.4.7 Endpoint Monitoring

● **INTERMEDIATE** **MANDATORY** ~10h *Prereq: IOC Analysis & Malware Basics*

Extends visibility down to individual hosts, where most attacker activity actually happens.

Covers: Sysmon event logging • Velociraptor artifact collection

Hands-on labs:

-
-

Resources:

- **Official Docs:** [Sysmon — Windows Sysinternals](#)
 - **Repository:** [Velociraptor Documentation](#)
-

5.4.8 Threat Intelligence

● **BEGINNER** **MANDATORY** ~8h Prereq: None

Enriches every investigation above with external context on who/what is behind an indicator.

Covers: VirusTotal enrichment • OSINT enrichment basics

Hands-on labs:

-

Resources:

- **Official Docs:** [VirusTotal](#)
-

5.5 Weekly Milestones

Week	Focus & Objective	Key Deliverable	Checkpoint
Wk 3	SOC operations model understood; SIEM (Wazuh/ELK) deployed and queried	Working SIEM with onboarded log sources	Lab review + quiz
Wk 4	ATT&CK mapping and Sigma rule authoring operational	Mini-Project 1: 5 custom detection rules	Mini-project demo
Wk 5	Full IR lifecycle + malware/IOC triage practiced	Mini-Project 2: End-to-end incident report	Mini-project demo + peer review
Wk 6	Endpoint telemetry and threat intel integrated into investigations	Capstone: SOC detection & investigation lab	Capstone showcase

5.6 Sample Daily Activities

Representative week (Week 4) — the same daily rhythm (concept → guided lab → independent practice) repeats each week with new topics.

Day	Concept Block (AM)	Guided Lab (PM)	Independent Practice / Output
Mon	ATT&CK tactics/techniques deep-dive	Map a sample attack narrative to ATT&CK	Navigator layer submitted for review
Tue	Threat-hunting hypothesis design	Hunt for T1059 across sample logs	Hunt findings write-up
Wed	Sigma rule syntax	Author a Sigma rule for a suspicious pattern	Rule converted & tested against SIEM
Thu	Detection tuning & false-positive reduction	Tune a noisy rule against sample data	Before/after FP-rate comparison
Fri	Week 4 review + mentor office hours	Mini-Project 1 build & polish	Mini-Project 1 live demo + weekly quiz

5.7 Mini-Projects

Mini-Project 1: Five Custom Detection Rules (end of Week 4)

Author 5 Sigma rules covering at least 3 different MITRE ATT&CK tactics, convert each to a working SIEM query, and validate each against sample or self-generated log data.

Deliverable: 5 Sigma rule files + SIEM query exports + a short validation write-up per rule (what it catches, why, tested false-positive rate).

Mini-Project 2: End-to-End Incident Report (end of Week 5)

Given a staged, multi-stage compromise in the lab environment, run the full IR lifecycle — detect via SIEM, contain, collect endpoint artifacts with Velociraptor, and produce a formal incident report.

Deliverable: Formal incident report (timeline, root cause, IOCs, remediation) + supporting SIEM/Velociraptor evidence exports.

5.8 Assessment Checkpoints

Checkpoint	Covers	Format	Weight
Weekly Labs (4 sets)	That week's cluster topics	Evidence export + mentor review	35%
Weekly Quiz (×4)	Concepts from that week	20 MCQ/short-answer, 30 min	15%
Mini-Project 1 & 2	Weeks 4–5 integrated skills	Deliverable + live walkthrough	25%
Capstone	Full track	Live demo + detection rules + IR report	25%

5.9 Capstone Project

Build a SOC Lab that Detects, Investigates & Reports Simulated Attacks

In pairs, interns stand up a mini-SOC (SIEM + endpoint telemetry + threat intel enrichment) monitoring a small lab network. A mentor (or automated attack simulation) then runs a multi-stage attack against the environment without prior warning, and interns must detect, investigate, and report it live.

Requirements:

- Working SIEM (Wazuh, ELK, or Splunk) ingesting logs from at least 3 sources.
- Minimum 8 custom detection rules (Sigma or native SIEM) mapped to ATT&CK.
- Sysmon deployed on at least one monitored endpoint.
- A documented incident-response playbook used during the live exercise.
- A professional incident report produced within 24 hours of the simulated attack.

Deliverables:

- SOC dashboard (live).
- Detection rule set with ATT&CK mapping.
- Incident-response playbook.
- Final incident report.
- 15-minute live demo replaying the detection-to-report timeline.

Evaluation: Scored against the Capstone Evaluation Rubric in Section 9.5 across detection coverage, investigation quality, response time, and report clarity.

5.10 Career Outcomes

This track prepares interns for entry-level roles such as: **SOC Analyst (L1/L2) · Security Analyst · Threat Detection Analyst · Incident Response Associate**

6. Track 3 — Cloud (OpenStack)

Deploy, operate, and automate a real private cloud on OpenStack — this specialization is 100% OpenStack, not AWS/Azure/GCP.

Field	Detail
Duration	4 weeks (Weeks 3–6)
Prerequisites	Phase 1 complete — especially Linux, Virtualization & Containers, and Networking Fundamentals
Estimated Study Hours	~115–125 hours across 4 weeks
Target Job Roles	Cloud / OpenStack Engineer · Cloud Infrastructure Associate · DevOps Engineer (Cloud) · Cloud Support Engineer

6.1 Learning Objectives

- Explain core cloud computing models (IaaS/PaaS/SaaS) and how OpenStack implements IaaS.
- Deploy OpenStack via both DevStack (single-node, dev) and Kolla-Ansible (containerized, production-style).
- Operate the core services — Keystone, Nova, Neutron, Glance, Cinder, Swift — from CLI, API, and Horizon.
- Configure tenant networking: provider/self-service networks, security groups, and floating IPs.
- Automate provisioning with Terraform and Ansible against the OpenStack API.
- Design for high availability across a multi-node deployment and monitor cloud health with Prometheus/Grafana.

6.2 Required Tools

Tool	Purpose
DevStack	Fast single-node OpenStack install for learning and development
Kolla-Ansible	Containerized, production-style multi-node OpenStack deployment
OpenStack CLI	Command-line control of every OpenStack service
Terraform	Infrastructure-as-Code provisioning against the OpenStack provider
Ansible	Configuration management and OpenStack automation
Prometheus	Metrics collection for cloud infrastructure health
Grafana	Dashboards and alerting on top of Prometheus

6.3 Track Roadmap

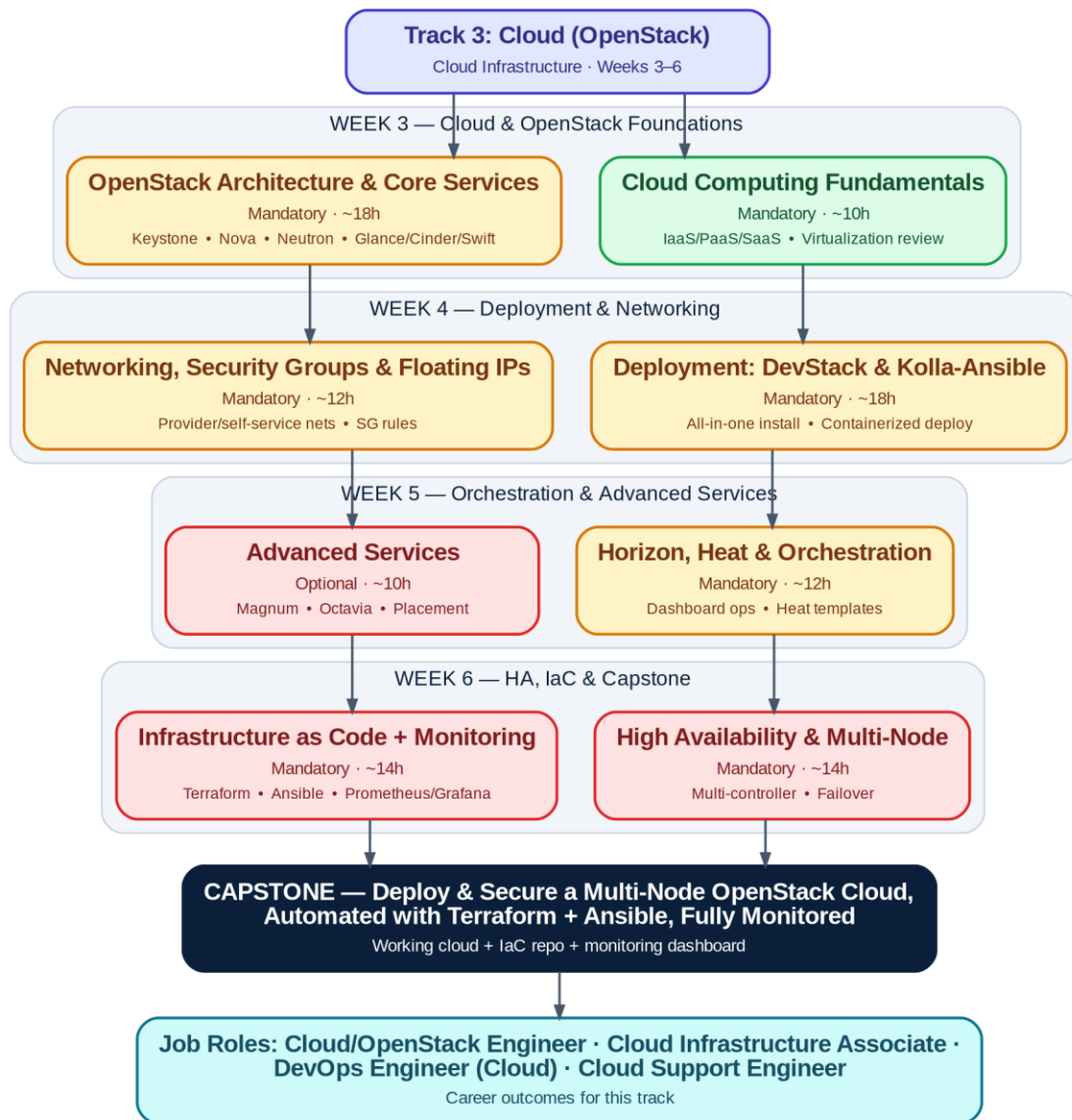


Figure 5 — Cloud (OpenStack) track roadmap: Weeks 3–6 from cloud fundamentals to a production-style multi-node capstone

6.4 Topic Breakdown

6.4.1 Cloud Computing Fundamentals

● **BEGINNER** **MANDATORY** ~10h *Prereq: Phase 1 Virtualization & Containers*

The vocabulary and service models every cloud engineer needs before touching OpenStack itself.

Covers: IaaS / PaaS / SaaS • Public / private / hybrid cloud • Virtualization review

Hands-on labs:

-

Resources:

- **Official Docs:** [OpenStack Documentation home](#)
- **Free Course:** [roadmap.sh — DevOps roadmap \(cloud fundamentals section\)](#)

6.4.2 OpenStack Architecture & Core Services

● **INTERMEDIATE** **MANDATORY** ~18h *Prereq: Cloud Computing Fundamentals*

The services that make up 'OpenStack' — identity, compute, network, image, and storage — and how they talk to each other.

Covers: Keystone (identity) • Nova (compute) • Neutron (network) • Glance (image) • Cinder (block storage) • Swift (object storage)

Hands-on labs:

-
-

Resources:

- **Official Docs:** [OpenStack Documentation home](#)
- **Free Course:** [OpenStack Install Guide](#)
- **Hands-on / Practice:** [OpenStack Client \(CLI\) Documentation](#)

6.4.3 Deployment: DevStack & Kolla-Ansible

● **INTERMEDIATE** **MANDATORY** ~18h *Prereq: OpenStack Architecture & Core Services*

Two deployment philosophies: DevStack for fast iteration, Kolla-Ansible for something closer to production.

Covers: DevStack all-in-one install • Kolla-Ansible containerized deploy • globals.yml configuration

Hands-on labs:

-
-

Resources:

- **Official Docs:** [DevStack Documentation](#)
- **Free Course:** [Kolla Ansible Documentation](#)
- **Repository:** [openstack/kolla-ansible \(GitHub\)](#)

6.4.4 Networking, Security Groups & Floating IPs

● **INTERMEDIATE** **MANDATORY** ~12h *Prereq: Deployment: DevStack & Kolla-Ansible*

Tenant networking is where most real-world OpenStack troubleshooting time goes — worth its own dedicated cluster.

Covers: Provider vs. self-service networks • Security group rules • Floating IP allocation & association

Hands-on labs:

-
-

Resources:

- **Official Docs:** [OpenStack Networking Guide](#)
- **Hands-on / Practice:** [OpenStack Client \(CLI\) Documentation](#)

6.4.5 Horizon, Heat & Orchestration

• **INTERMEDIATE MANDATORY** ~12h *Prereq: Networking, Security Groups & Floating IPs*

The dashboard operators actually click through daily, plus template-driven orchestration for repeatable stacks.

Covers: Horizon dashboard operations • Heat Orchestration Templates (HOT)

Hands-on labs:

-
-

Resources:

- **Official Docs:** [OpenStack Documentation home](#)

6.4.6 Advanced Services

• **ADVANCED OPTIONAL** ~10h *Prereq: Horizon, Heat & Orchestration*

Optional deep cuts for interns who want exposure to container-on-OpenStack and load-balancing services.

Covers: Magnum (container orchestration) • Octavia (load balancing as a service) • Placement (resource scheduling)

Hands-on labs:

-

Resources:

- **Official Docs:** [OpenStack Documentation home](#)

6.4.7 High Availability & Multi-Node

• **ADVANCED MANDATORY** ~14h *Prereq: Deployment: DevStack & Kolla-Ansible*

Single-node clouds don't survive a reboot gracefully — this is where the deployment becomes genuinely production-minded.

Covers: Multi-controller design • Failover behavior • Backup basics

Hands-on labs:

-

Resources:

- **Official Docs:** [Kolla Ansible Documentation](#)

6.4.8 Infrastructure as Code + Monitoring

- **ADVANCED MANDATORY** ~14h Prereq: High Availability & Multi-Node

Ties the whole track together: nothing gets clicked by hand, and everything is observable.

Covers: Terraform (OpenStack provider) • Ansible for OpenStack • Prometheus + Grafana monitoring

Hands-on labs:

-
-

Resources:

- **Official Docs:** [HashiCorp Terraform Documentation](#)
 - **Free Course:** [Terraform OpenStack Provider Docs](#)
 - **Hands-on / Practice:** [Ansible Documentation](#)
 - **Repository:** [Prometheus Documentation](#)
-

6.5 Weekly Milestones

Week	Focus & Objective	Key Deliverable	Checkpoint
Wk 3	Cloud fundamentals understood; OpenStack architecture mapped and a single-node cloud running via DevStack	Working DevStack all-in-one deployment	Lab review + quiz
Wk 4	Kolla-Ansible deployment and tenant networking operational	Mini-Project 1: Networked, access-controlled DevStack tenant	Mini-project demo
Wk 5	Horizon/Heat orchestration and multi-node HA design practiced	Mini-Project 2: Heat-templated stack on a multi-node cloud	Mini-project demo + peer review
Wk 6	IaC (Terraform/Ansible) and monitoring layered on top of the cloud	Capstone: production-style multi-node OpenStack cloud	Capstone showcase

6.6 Sample Daily Activities

Representative week (Week 4) — the same daily rhythm (concept → guided lab → independent practice) repeats each week with new topics.

Day	Concept Block (AM)	Guided Lab (PM)	Independent Practice / Output
Mon	Kolla-Ansible architecture & globals.yml	Deploy Kolla-Ansible all-in-one	Compare operational experience vs. DevStack (write-up)
Tue	Kolla-Ansible service selection	Enable an additional service (e.g., Cinder) and redeploy	Verify new service via CLI
Wed	Tenant networking deep-dive	Build an isolated network + router + floating IP	Verify external connectivity to instance
Thu	Security groups & least privilege	Write and apply restrictive security group rules	Port-scan verification + write-up
Fri	Week 4 review + mentor office hours	Mini-Project 1 build & polish	Mini-Project 1 live demo + weekly quiz

6.7 Mini-Projects

Mini-Project 1: Networked, Access-Controlled Tenant (end of Week 4)

On your Kolla-Ansible or DevStack cloud, build a tenant environment with an isolated network, a router to the external network, 2+ instances, least-privilege security groups, and a floating IP for external access.

Deliverable: OpenStack CLI command history / script + network diagram + short write-up of the security group logic.

Mini-Project 2: Heat-Templated Stack on a Multi-Node Cloud (end of Week 5)

Write a Heat Orchestration Template that provisions a full stack (network, security group, instance) in one command, and deploy it against a multi-node Kolla-Ansible cloud.

Deliverable: Heat template (.yaml) + successful stack-create output + Horizon screenshot of the running stack.

6.8 Assessment Checkpoints

Checkpoint	Covers	Format	Weight
Weekly Labs (4 sets)	That week's cluster topics	Command history / config export, reviewed by mentor	35%
Weekly Quiz (×4)	Concepts from that week	20 MCQ/short-answer, 30 min	15%
Mini-Project 1 & 2	Weeks 4–5 integrated skills	Working deployment + live walkthrough	25%
Capstone	Full track	Live demo + IaC repo + monitoring dashboard	25%

6.9 Capstone Project

Deploy & Secure a Production-Style Multi-Node OpenStack Cloud

In pairs, interns deploy a multi-node OpenStack cloud (minimum controller + compute + storage roles) using Kolla-Ansible, secure it with proper tenant networking and security groups, automate provisioning end-to-end with Terraform and Ansible, and monitor the whole stack with Prometheus and Grafana.

Requirements:

- Minimum 3-node Kolla-Ansible deployment with distinct controller/compute/storage roles.
- At least one tenant network with security groups and a floating IP pool.
- A Terraform configuration that provisions the full workload (network + instances) from scratch.
- An Ansible playbook automating at least one post-provisioning configuration task.
- A Prometheus + Grafana dashboard showing cloud and instance health.

Deliverables:

- Working multi-node cloud (demoed live).
- Terraform + Ansible repository (IaC).
- Monitoring dashboard.
- Architecture diagram.
- 15-minute live demo including a simulated node/service failure.

Evaluation: Scored against the Capstone Evaluation Rubric in Section 9.5 across deployment correctness, security configuration, automation quality, monitoring coverage, and documentation.

6.10 Career Outcomes

This track prepares interns for entry-level roles such as: **Cloud / OpenStack Engineer** • **Cloud Infrastructure Associate** • **DevOps Engineer (Cloud)** • **Cloud Support Engineer**

7. Track 4 — Cybersecurity (Red Team / Pentesting)

Think like an attacker, under a signed scope and rules of engagement, to find the holes defenders need to fix first.

Field	Detail
Duration	4 weeks (Weeks 3–6)
Prerequisites	Phase 1 complete — especially Cybersecurity Fundamentals, Linux, and Networking
Estimated Study Hours	~120–130 hours across 4 weeks
Target Job Roles	Junior Penetration Tester · Red Team Associate · Vulnerability Assessment Analyst · Ethical Hacker (Entry-Level)

7.1 Learning Objectives

- Perform structured reconnaissance and OSINT gathering against an in-scope target.
- Scan, enumerate, and fingerprint services with Nmap and interpret results correctly.
- Identify and exploit OWASP Top 10 web vulnerabilities using Burp Suite.
- Explain core Active Directory attack paths and map them with BloodHound.
- Escalate privileges on both Linux and Windows targets using structured methodology, not guesswork.
- Use Metasploit and related frameworks responsibly within a defined scope, and write a client-ready penetration-test report.

7.2 Required Tools

Tool	Purpose
Kali Linux	Primary penetration-testing operating system and tool distribution
Burp Suite	Web application proxy, scanner, and exploitation toolkit
Metasploit	Exploitation framework for validated, in-scope exploitation
BloodHound	Active Directory attack-path mapping and analysis
NetExec (formerly CrackMapExec)	Network/AD enumeration and credential-testing swiss-army knife
Gobuster	Directory, file, and DNS subdomain brute-forcing
Hydra	Online password/brute-force attacks against network services
John the Ripper	Offline password-hash cracking

7.3 Track Roadmap

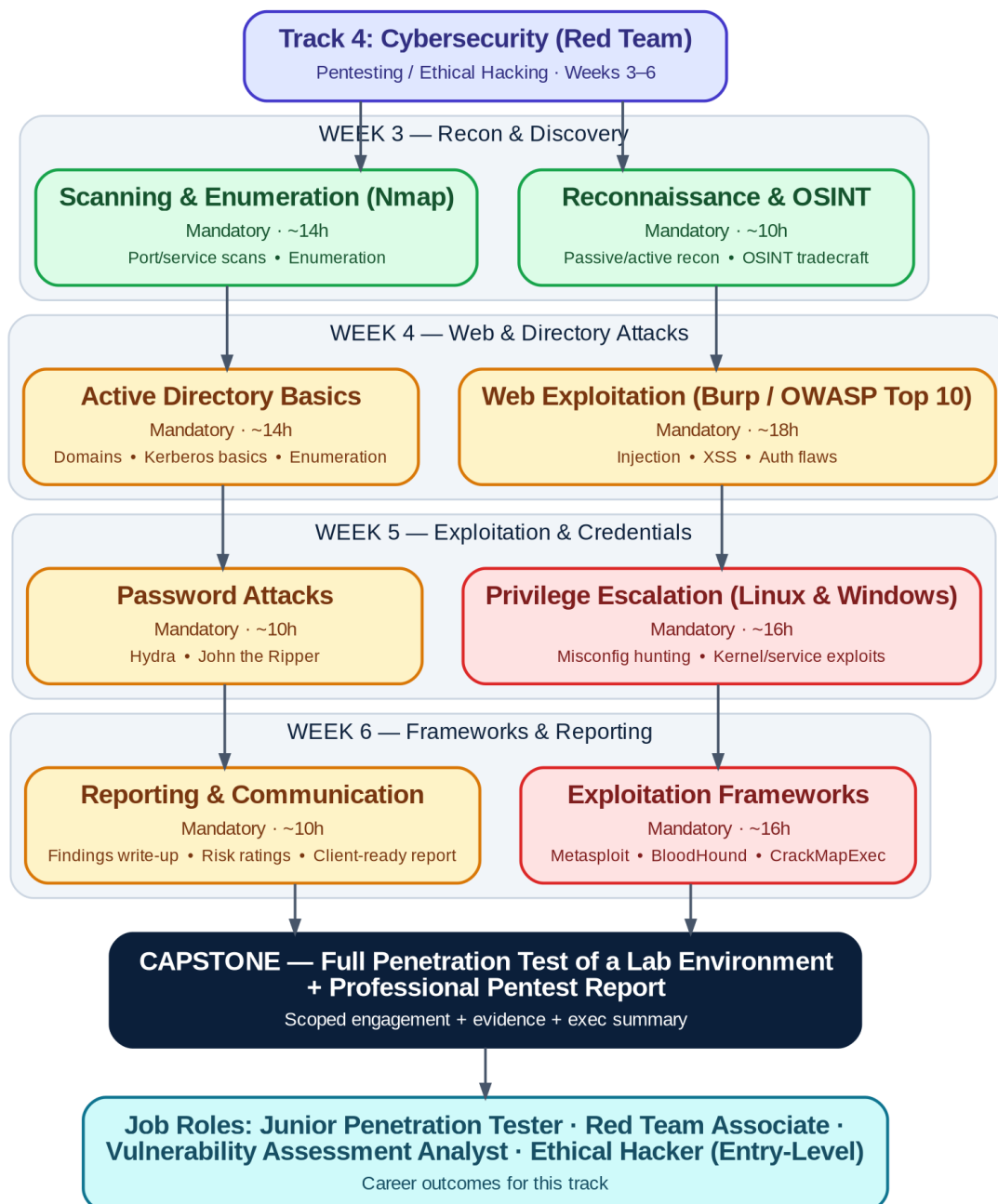


Figure 6 — Red Team track roadmap: Weeks 3–6 from reconnaissance to a full penetration-test capstone

7.4 Topic Breakdown

7.4.1 Reconnaissance & OSINT

● **BEGINNER** **MANDATORY** ~10h *Prereq: Phase 1 Cybersecurity Fundamentals*

Every real engagement starts here — understanding the target before touching it.

Covers: Passive vs. active reconnaissance • OSINT tradecraft • Scope discipline

Hands-on labs:

-
-

Resources:

- **Official Docs:** [OWASP Testing Guide](#)
 - **Free Course:** [TryHackMe](#)
 - **Hands-on / Practice:** [OSINT Framework](#)
-

7.4.2 Scanning & Enumeration (Nmap)

● **BEGINNER** **MANDATORY** ~14h *Prereq: Reconnaissance & OSINT*

Turns a target IP range into a concrete list of services and versions worth investigating.

Covers: Port/service scanning • Version & OS fingerprinting • Nmap scripting engine (NSE)

Hands-on labs:

-
-

Resources:

- **Official Docs:** [Nmap Reference Guide](#)
 - **Hands-on / Practice:** [HackTheBox Academy — course catalogue](#)
-

7.4.3 Web Exploitation (Burp Suite / OWASP Top 10)

● **INTERMEDIATE** **MANDATORY** ~18h *Prereq: Scanning & Enumeration (Nmap)*

The highest-volume vulnerability class in real engagements, and where Burp Suite becomes second nature.

Covers: Injection (SQLi, command injection) • Cross-Site Scripting (XSS) • Authentication & session flaws • OWASP Top 10 mapping

Hands-on labs:

-
-

Resources:

- **Official Docs:** [OWASP Top 10](#)
 - **Free Course:** [PortSwigger Web Security Academy \(free\)](#)
 - **Hands-on / Practice:** [PortSwigger Web Security Academy \(free\)](#)
-

7.4.4 Active Directory Basics

● **INTERMEDIATE** **MANDATORY** ~14h *Prereq: Web Exploitation*

Most enterprise internal engagements are, in practice, an Active Directory engagement.

Covers: AD domains, trusts & Kerberos basics • Enumeration with NetExec/BloodHound

Hands-on labs:

-
-

Resources:

- **Official Docs:** [BloodHound Documentation](#)
 - **Hands-on / Practice:** [HackTheBox Academy — Penetration Tester path](#)
 - **Repository:** [BloodHound \(GitHub\)](#)
-

7.4.5 Privilege Escalation (Linux & Windows)

• **ADVANCED MANDATORY** ~16h *Prereq: Active Directory Basics*

Getting in is half the job — privilege escalation is what turns initial access into real impact.

Covers: Linux misconfiguration hunting (SUID, cron, PATH) • Windows service/kernel misconfigurations

Hands-on labs:

-
-

Resources:

- **Official Docs:** [GTFOBins \(Linux living-off-the-land binaries\)](#)
 - **Hands-on / Practice:** [HackTheBox Academy — Penetration Tester path](#)
-

7.4.6 Password Attacks

• **INTERMEDIATE MANDATORY** ~10h *Prereq: Privilege Escalation*

Weak or reused credentials remain one of the most reliable ways into a network.

Covers: Online brute-force (Hydra) • Offline hash cracking (John the Ripper) • Password-spraying discipline

Hands-on labs:

-
-

Resources:

- **Official Docs:** [John the Ripper — Openwall](#)
 - **Repository:** [THC-Hydra \(GitHub\)](#)
-

7.4.7 Exploitation Frameworks

• **ADVANCED MANDATORY** ~16h *Prereq: Password Attacks*

Brings reconnaissance, vulnerability identification, and privilege escalation together into a single guided workflow.

Covers: Metasploit modules & workflow • BloodHound-guided AD attacks • NetExec (formerly CrackMapExec) for lateral movement

Hands-on labs:

-
-

Resources:

- **Official Docs:** [Metasploit Documentation](#)
 - **Repository:** [NetExec \(GitHub\)](#) — formerly CrackMapExec
-

7.4.8 Reporting & Communication

● **INTERMEDIATE MANDATORY** ~10h *Prereq: Exploitation Frameworks*

A finding nobody can act on isn't a finding — this is the skill that makes the rest of the track employable.

Covers: Findings write-up structure • Risk ratings (CVSS) • Client-ready executive summaries

Hands-on labs:

-

Resources:

- **Official Docs:** [Penetration Testing Execution Standard \(PTES\)](#)
 - **Free Course:** [OWASP Testing Guide](#)
-

7.5 Weekly Milestones

Week	Focus & Objective	Key Deliverable	Checkpoint
Wk 3	Recon/OSINT and scanning/enumeration methodology established	Full service inventory + OSINT profile for lab target	Lab review + quiz
Wk 4	Web exploitation and AD enumeration operational	Mini-Project 1: Web app compromise report	Mini-project demo
Wk 5	Privilege escalation and password attacks practiced on Linux & Windows	Mini-Project 2: Full privesc chain, both OS	Mini-project demo + peer review
Wk 6	Exploitation frameworks and reporting brought together	Capstone: full penetration test + report	Capstone showcase

7.6 Sample Daily Activities

Representative week (Week 4) — the same daily rhythm (concept → guided lab → independent practice) repeats each week with new topics.

Day	Concept Block (AM)	Guided Lab (PM)	Independent Practice / Output
Mon	OWASP Top 10: injection & XSS theory	Exploit an injection flaw with Burp Suite	Documented exploitation steps + screenshots
Tue	Authentication/session flaw theory	Complete 3 PortSwigger Academy labs	Lab completion certificates + notes
Wed	Active Directory concepts & Kerberos basics	Enumerate lab AD with NetExec + BloodHound	BloodHound attack-path screenshot + explanation
Thu	Report-writing standards (CVSS, risk framing)	Draft findings section for Week 4 vulnerabilities	Peer-reviewed findings write-up
Fri	Week 4 review + mentor office hours	Mini-Project 1 build & polish	Mini-Project 1 live demo + weekly quiz

7.7 Mini-Projects

Mini-Project 1: Web Application Compromise Report (end of Week 4)

Against a deliberately vulnerable lab web application, identify and exploit at least 3 distinct OWASP Top 10 vulnerabilities using Burp Suite, then write a findings report for each with risk rating and remediation advice.

Deliverable: Findings report (3+ vulnerabilities) + Burp Suite evidence exports + retest notes.

Mini-Project 2: Full Privilege-Escalation Chain — Linux & Windows (end of Week 5)

On both a Linux and a Windows lab target, gain initial low-privilege access and escalate to full administrative control, documenting every step and technique used.

Deliverable: Two escalation write-ups (one per OS) with commands, screenshots, and technique explanations.

7.8 Assessment Checkpoints

Checkpoint	Covers	Format	Weight
Weekly Labs (4 sets)	That week's cluster topics	Evidence export + mentor review	35%
Weekly Quiz (×4)	Concepts from that week	20 MCQ/short-answer, 30 min	15%
Mini-Project 1 & 2	Weeks 4–5 integrated skills	Findings report + live walkthrough	25%
Capstone	Full track	Live demo + full pentest report	25%

7.9 Capstone Project

Full Penetration Test of a Lab Environment + Professional Report

Working individually under a signed mock "Rules of Engagement," each intern performs a complete black-box-to-grey-box penetration test of a multi-host lab environment (mixed Linux/Windows, at least one web application, basic AD) and delivers a professional report as if for a paying client.

Requirements:

- Full recon-to-report methodology followed and documented (recon → scanning → exploitation → post-exploitation → reporting).
- At least one web-application vulnerability and one host/OS-level compromise, each fully chained to demonstrated impact.
- Every action stays within the signed scope document — no out-of-scope testing.
- Findings rated with CVSS and prioritized by business risk, not just technical severity.
- A professional report including executive summary, technical findings, and remediation roadmap.

Deliverables:

- Signed scope / rules-of-engagement document.
- Full technical findings log with evidence.
- Professional penetration-test report (executive summary + technical detail).
- 15-minute live demo of the primary attack chain.

Evaluation: Scored against the Capstone Evaluation Rubric in Section 9.5 across methodology rigor, technical depth, scope discipline, and report quality.

7.10 Career Outcomes

This track prepares interns for entry-level roles such as: **Junior Penetration Tester** · **Red Team Associate** · **Vulnerability Assessment Analyst** · **Ethical Hacker (Entry-Level)**

8. Track 5 — R&D (Research & Development)

Pick an emerging-technology domain, run it like a real R&D sprint, and ship a working prototype — not just a tutorial follow-along.

Field	Detail
Duration	4 weeks (Weeks 3–6)
Prerequisites	Phase 1 complete — especially Python Programming; domain-specific prerequisites noted per domain below
Estimated Study Hours	~110–120 hours across 4 weeks (domain-dependent)
Target Job Roles	Junior AI/ML Engineer · Mobile App Developer (Android/iOS) · DevSecOps Engineer · QA/Automation Engineer · R&D / Innovation Associate

8.1 Learning Objectives

- Scope a research/build problem properly: feasibility check, success criteria, and a realistic 4-week plan.
- Reach working proficiency in one chosen emerging-technology domain (AI, Android, iOS, DevSecOps, or Automation).
- Apply engineering rigor — version control, testing, documentation — to exploratory, less-defined R&D work.
- Build a working prototype that demonstrates the chosen domain's core skills end-to-end.
- Communicate technical findings and a working demo clearly to a non-specialist audience.

8.2 Required Tools

Tool	Purpose
Python + LangChain / Hugging Face	AI/LLM domain: model access, RAG pipelines, agents
Android Studio + Kotlin + Jetpack Compose	Android domain: native app development
Xcode + Swift + SwiftUI	iOS domain: native app development
GitHub Actions / Jenkins + Docker + Kubernetes + Trivy/SonarQube	DevSecOps domain: pipelines with security scanning built in
Selenium / Playwright + PowerShell	Automation domain: test and process automation

8.3 Track Roadmap

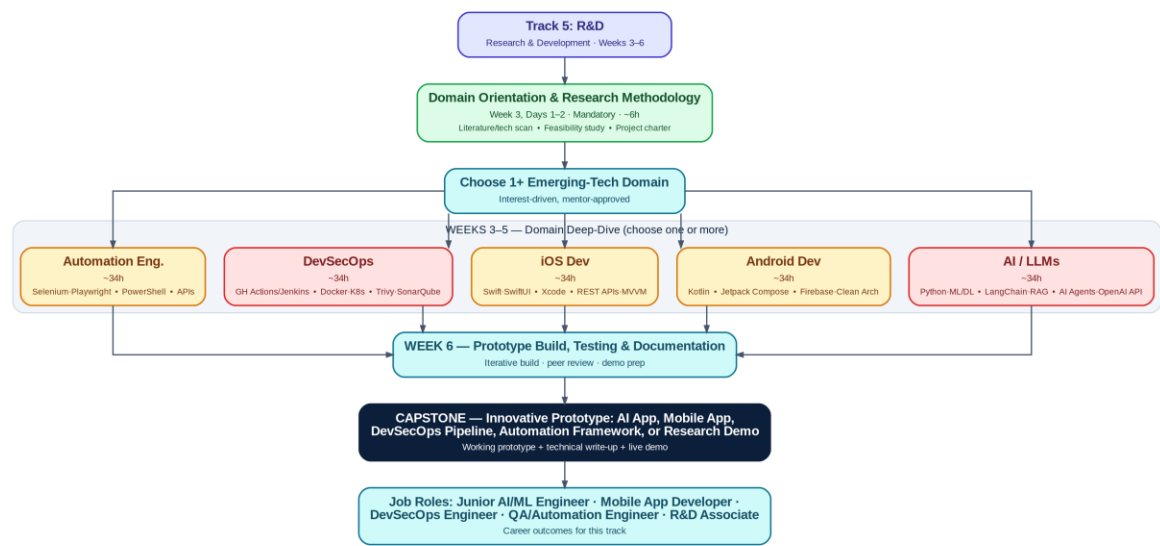


Figure 7 — R&D track roadmap: shared orientation, a chosen domain deep-dive, then a converged prototype capstone

8.4 Topic Breakdown

8.4.1 Domain Orientation & Research Methodology

• **BEGINNER MANDATORY** ~6h *Prereq: Phase 1 Python Programming*

Before picking a lane, every intern learns to scope a technical problem like a real R&D engineer would.

Covers: Literature / technology scan • Feasibility study • Writing a one-page project charter

Hands-on labs:

-
-

Resources:

- **Official Docs:** [roadmap.sh — role-based roadmaps \(browse all domains\)](#)

8.4.2 Domain: Artificial Intelligence & LLMs

• **ADVANCED OPTIONAL** ~34h *Prereq: Domain Orientation (select 1+ domains)*

Covers classical ML basics fast, then spends most of its time on the LLM application stack interns are most likely to use on the job.

Covers: Advanced Python • Machine Learning / Deep Learning basics • LLMs & Prompt Engineering • LangChain • RAG (Retrieval-Augmented Generation) • AI Agents • OpenAI-compatible APIs

Hands-on labs:

-
-
-

Resources:

- **Official Docs:** [OpenAI API Documentation](#)
- **Free Course:** [roadmap.sh — AI Engineer roadmap](#)
- **Hands-on / Practice:** [Hugging Face — models & datasets](#)
- **Repository:** [LangChain Documentation](#)

8.4.3 Domain: Android Development

• **INTERMEDIATE OPTIONAL** ~34h *Prereq: Domain Orientation (select 1+ domains)*

Modern, Kotlin-first native Android development using the current recommended toolkit.

Covers: Kotlin • Android Studio • Jetpack Compose • Firebase • REST APIs • Clean Architecture basics

Hands-on labs:

-
-

Resources:

- **Official Docs:** [Android Developers Documentation](#)
- **Free Course:** [roadmap.sh — Android roadmap](#)
- **Hands-on / Practice:** [Kotlin Documentation](#)
- **Repository:** [Firebase Documentation](#)

8.4.4 Domain: iOS Development

● **INTERMEDIATE** OPTIONAL ~34h *Prereq: Domain Orientation (select 1+ domains)*

Modern, SwiftUI-first native iOS development using Apple's current recommended toolkit.

Covers: Swift • SwiftUI • Xcode • REST APIs • MVVM architecture

Hands-on labs:

-
-

Resources:

- **Official Docs:** [Apple Developer Documentation](#)
 - **Free Course:** [Apple — Develop apps for iOS \(SwiftUI tutorials\)](#)
 - **Hands-on / Practice:** [Apple Developer Documentation — Xcode](#)
-

8.4.5 Domain: DevSecOps

● **ADVANCED** OPTIONAL ~34h *Prereq: Domain Orientation (select 1+ domains); Phase 1 Virtualization & Containers*

Builds security scanning into the pipeline itself, rather than bolting it on afterward.

Covers: GitHub Actions / Jenkins • Docker • Kubernetes basics • Security scanning — Trivy, SonarQube

Hands-on labs:

-
-

Resources:

- **Official Docs:** [roadmap.sh — DevSecOps roadmap](#)
 - **Free Course:** [GitHub Actions Documentation](#)
 - **Hands-on / Practice:** [Kubernetes Documentation](#)
 - **Repository:** [Trivy \(GitHub\)](#)
-

8.4.6 Domain: Automation Engineering

● **INTERMEDIATE** OPTIONAL ~34h *Prereq: Domain Orientation (select 1+ domains)*

Builds the test-automation and process-automation skills that keep QA and IT-ops teams sane.

Covers: Selenium • Playwright • PowerShell • API test automation

Hands-on labs:

-
-

Resources:

- **Official Docs:** [Selenium Documentation](#)
 - **Free Course:** [Playwright Documentation](#)
 - **Hands-on / Practice:** [PowerShell Documentation \(Microsoft Learn\)](#)
-

8.5 Weekly Milestones

Week	Focus & Objective	Key Deliverable	Checkpoint
Wk 3	Domain chosen; feasibility study and project charter approved by mentor	Project charter + environment set up for chosen domain	Charter review
Wk 4	Core domain skills built through guided labs	Mini-Project 1: smallest working version of the prototype	Mini-project demo
Wk 5	Prototype expanded to full planned feature set	Mini-Project 2: feature-complete prototype (pre-polish)	Mini-project demo + peer review
Wk 6	Prototype hardened, tested, documented, and rehearsed for demo	Capstone: finished prototype + technical write-up	Capstone showcase

8.6 Sample Daily Activities

Representative week (Week 4) — the same daily rhythm (concept → guided lab → independent practice) repeats each week with new topics.

Day	Concept Block (AM)	Guided Lab (PM)	Independent Practice / Output
Mon	Domain concept block (varies by chosen domain)	Guided lab implementing that concept	Code committed with tests where applicable
Tue	Domain concept block (continued)	Guided lab implementing that concept	Code committed with tests where applicable
Wed	Integration concept (tying prior labs together)	Build the connecting piece in your prototype	Working integrated slice demoed to a peer
Thu	Code review norms & documentation practice	Refactor + document the week's work	Updated README / technical notes
Fri	Week 4 review + mentor office hours	Mini-Project 1 build & polish	Mini-Project 1 live demo + weekly quiz

8.7 Mini-Projects

Mini-Project 1: Smallest Working Prototype (end of Week 4)

Build the smallest possible end-to-end version of your capstone idea — every layer present (e.g., for AI: ingest → retrieve → generate; for Android/iOS: UI → API call → display; for DevSecOps: commit → build → scan → deploy) even if each layer is minimal.

Deliverable: Working code repository + a 3-minute walkthrough demo.

Mini-Project 2: Feature-Complete Prototype (end of Week 5)

Expand the Week 4 prototype to the full feature set defined in your Week 3 project charter, without yet focusing on polish, edge cases, or performance.

Deliverable: Updated repository + a written list of what's complete vs. what remains for capstone polish.

8.8 Assessment Checkpoints

Checkpoint	Covers	Format	Weight
Weekly Labs / Progress Check (4 sets)	That week's domain progress	Repo review + mentor check-in	30%
Charter & Feasibility Review	Week 3 planning quality	Written charter + mentor approval	10%
Mini-Project 1 & 2	Weeks 4–5 integrated skills	Working prototype + live walkthrough	30%
Capstone	Full track	Live demo + technical write-up	30%

8.9 Capstone Project

Innovative Prototype — AI App, Mobile App, DevSecOps Pipeline, Automation Framework, or Research Demo

Individually or in pairs, interns deliver a finished, working prototype in their chosen domain that goes meaningfully beyond a tutorial clone — solving a real (even if small) problem, with the engineering rigor (tests, docs, version control) expected of professional R&D work.

Requirements:

- Prototype implements the full feature set committed to in the Week 3 project charter.
- Code is version-controlled with a clean commit history and a professional README.
- At least basic automated tests or a documented manual test plan.
- A short technical write-up explaining architecture, key decisions, and known limitations.
- A working, demoable build — not slides describing what it would do.

Deliverables:

- Source code repository.
- Technical write-up (architecture + decisions + limitations).
- Test evidence (automated or documented manual).
- 15-minute live demo.

Evaluation: Scored against the Capstone Evaluation Rubric in Section 9.5 across functionality, technical depth, code/engineering quality, and communication.

8.10 Career Outcomes

This track prepares interns for entry-level roles such as: **Junior AI/ML Engineer** · **Mobile App Developer (Android/iOS)** · **DevSecOps Engineer** · **QA/Automation Engineer** · **R&D / Innovation Associate**

9. Intern KPI & Evaluation Templates

This section provides the complete set of evaluation templates used to track intern performance across the program — from daily labs through the final capstone. Templates are designed to be copied directly into a spreadsheet or printed for in-person mentor reviews. All templates use the same 1–5 rating scale (defined in Section 9.10) so scores are comparable across weeks and tracks.

9.1 KPI Framework Overview

Every intern is evaluated on a continuous, multi-layered cadence rather than a single end-of-internship test. Each layer feeds into the Final Internship Performance Scorecard (Section 9.7):

- **Daily:** Lab completion, pass/revise checklist — feeds the Weekly Scorecard's Technical Skill Application score.
- **Weekly:** Weekly Intern KPI Scorecard (Section 9.2) + track-specific technical add-ons (Section 9.3), completed by the assigned mentor.
- **Milestone:** Mini-Project Evaluation Rubric (Section 9.4) at the end of Weeks 2, 4, and 5.
- **Mid-Point:** Mid-Internship Review Form (Section 9.6), end of Week 4.
- **Final:** Capstone Evaluation Rubric (Section 9.5) + Final Performance Scorecard (Section 9.7), Week 6.

Overall weighting toward the Final Internship Performance Scorecard:

Component	Weight
Daily Labs & Attendance (rolled up weekly)	15%
Weekly Quizzes	10%
Weekly KPI Scorecards — professionalism & soft skills	15%
Mini-Projects (Phase 1 + track-level)	20%
Capstone Project	30%
Mentor & Peer Feedback	10%

9.2 Weekly Intern KPI Scorecard (Template)

To be completed by the assigned mentor at the end of each week, ideally in a short 1:1 with the intern. Takes about 10 minutes.

Intern Name:		Track:	
Week #:	1-2	Mentor:	
Date:		Overall Weighted Score:	

Category	What to Look For	Score (1–5)	Weight	Weighted	Mentor Notes
Attendance & Punctuality	Present and on time for sessions and stand-ups		10%		
Technical Skill Application	Accuracy and quality of that week's lab work		25%		
Problem-Solving & Initiative	Works through blockers before escalating; proposes ideas		15%		
Documentation Quality	READMEs, comments, configs are clear and complete		10%		
Communication	Clear written updates and verbal explanations of work		10%		
Collaboration / Teamwork	Works well in pair/group labs; helps peers		10%		
Professionalism	Conduct, reliability, receptiveness to feedback		10%		
Growth vs. Previous Week	Visible improvement on prior week's feedback		10%		

Weighted = Score × Weight. Sum the Weighted column and multiply by 20 to express the Overall Weighted Score as a percentage (since $5 \times 20 = 100$).

9.3 Track-Specific Technical KPI Add-ons (Templates)

Layered on top of the "Technical Skill Application" row in Section 9.2, these track-specific metrics keep evaluation concrete and job-relevant rather than purely subjective.

Track 1 — NOC

Metric	Definition	Score (1–5)
Configuration accuracy	Config matches spec (VLAN/ACL/OSPF correctness) on first review	
Mean time to diagnose (MTTD-lab)	Minutes to correctly diagnose a mentor-injected fault	
Monitoring dashboard completeness	% of required devices/metrics represented with working alerts	
Network uptime maintained	% uptime sustained across lab exercises under fault injection	

Track 2 — SOC

Metric	Definition	Score (1–5)
Detection rule quality	True-positive rate of authored Sigma/SIEM rules against test data	
Mean time to detect (MTTD-sim)	Minutes from simulated attack start to first correct detection	
Mean time to respond (MTTR-sim)	Minutes from detection to containment action in tabletop exercises	
Incident report quality	Completeness & clarity of timeline, IOCs, and root cause	

Track 3 — Cloud (OpenStack)

Metric	Definition	Score (1–5)
Deployment success rate	% of deployments succeeding without manual rework	
IaC code quality	Terraform/Ansible passes lint checks and is idempotent on re-run	
HA resilience achieved	Service stays available through a simulated node failure (Y/N + notes)	
Runbook / documentation completeness	Deployment and recovery steps fully reproducible by a peer	

Track 4 — Red Team

Metric	Definition	Score (1–5)
Valid findings per engagement	Count of confirmed, reproducible vulnerabilities found	
Exploitation success within scope	% of identified vulnerabilities successfully (and safely) exploited	

Metric	Definition	Score (1–5)
Scope discipline	Zero out-of-scope actions — pass/fail gate, any failure reviewed immediately	
Report professionalism	Findings are reproducible, risk-rated, and client-ready	

Track 5 — R&D

Metric	Definition	Score (1–5)
Prototype functionality completeness	% of charter-committed features working end-to-end	
Code / test quality	Tests present (or documented manual test plan) and passing	
Innovation & problem-fit	Solution genuinely addresses the stated problem, not a tutorial clone	
Technical documentation clarity	A peer could understand the architecture from the write-up alone	

9.4 Mini-Project Evaluation Rubric (Template)

Reused for the Phase 1 mini-project and every track-level mini-project — just update the project title and deliverables reviewed.

Project Title:		Intern(s):	
Track / Phase:		Reviewer:	

Criterion	5 – Excellent	3 – Satisfactory	1 – Needs Work	Weight	Score
Functionality / Correctness	Fully working, handles edge cases	Works for the main use case	Partially working or broken	30%	
Technical Approach	Efficient, well-reasoned design choices	Reasonable, workable approach	Approach is unclear or fragile	20%	
Documentation	Clear, complete, easy to follow	Present but incomplete	Missing or unclear	20%	
Code / Config Quality	Clean, organized, follows conventions	Functional but messy	Hard to follow or maintain	15%	
Presentation / Demo	Confident, clear, answers questions well	Adequate walkthrough	Unclear or unprepared	15%	

9.5 Capstone Project Evaluation Rubric (Template)

Used for all 5 tracks' capstones. Track-specific evaluators should interpret each criterion through their track's lens (e.g., "Security & Best Practices" means firewall/ACL hygiene for NOC, detection coverage for SOC, secure tenant config for Cloud, etc.).

Intern(s):		Track:	
Capstone Title:		Date:	
Evaluator(s):		Final Score (/100):	

Criterion	Description	Weight	Score (1–5)	Weighted (/20)
Technical Complexity & Completeness	Scope matches or exceeds capstone requirements; all required components present	25%		
Correctness / Functionality	The system/deliverable works as intended, demonstrated live	20%		
Security & Best Practices	Track-appropriate security/quality hygiene applied throughout	15%		
Documentation & Reproducibility	A peer could rebuild or understand the work from the docs alone	15%		
Live Demo & Communication	Clear, confident walkthrough; handles follow-up questions well	15%		
Professionalism & Scope Discipline	Delivered on time, within scope, with professional conduct throughout	10%		

Scoring: Weighted (/20) = Score (1–5) × Weight × 100. Sum all Weighted values for the Final Score out of 100.

9.6 Mid-Internship Review Form (End of Week 4)

A structured checkpoint conversation between mentor and intern, roughly halfway through specialization.

Intern Name:		Track:	
Mentor:		Date:	

Average Weekly KPI Score so far (Weeks 3–4): _____ / Mini-Project 1 Score: _____

Key strengths observed so far:

Areas for growth before the capstone:

Is the intern on track for capstone readiness? ☐ On track ☐ On track with support ☐ At risk (escalate)

Specific action plan for Weeks 5–6:

Intern's own comments / requested support:

Mentor signature: _____ Intern signature: _____

9.7 Final Internship Performance Scorecard & Certificate Eligibility

Completed by the lead mentor in the final days of Week 6, after the capstone showcase.

Intern Name:		Track:	
Cohort Dates:		Evaluator:	

Component	Weight	Raw Score	Weighted
Daily Labs & Attendance (avg.)	15%		
Weekly Quizzes (avg.)	10%		
Weekly KPI Scorecards (avg.)	15%		
Mini-Projects (avg., Phase 1 + track)	20%		
Capstone Project	30%		
Mentor & Peer Feedback	10%		
TOTAL	100%		

Final Rating (see Section 9.10 scale): _____

Certificate of Completion issued? ☐ Yes ☐ No — see notes below

Notes / standout achievements / recommendation for future roles:

Certificate Eligibility Threshold

- Overall weighted score of 70% or higher, AND
- Capstone project fully delivered and demoed (not a placeholder submission), AND
- No unresolved professional-conduct flags from any Weekly KPI Scorecard or Mid-Internship Review.

9.8 Intern Self-Assessment

Name:		Week #:	1-2
-------	--	---------	-----

What I learned this week that I'm most confident in:

What still feels shaky or unclear:

One thing I did well this week:

One thing I'd do differently:

Support I'd like from my mentor next week:

Self-rated confidence (1 = not confident, 5 = very confident):

Skill Area	1	2	3	4	5
This week's core technical topic	☐	☐	☐	☐	☐
Tools used this week	☐	☐	☐	☐	☐
Communicating my work to others	☐	☐	☐	☐	☐

9.9 Mentor / Supervisor Weekly Feedback Template

A short structured narrative to accompany the Weekly KPI Scorecard — this is what actually gets read and remembered by the intern.

Intern:		Week #:	
Mentor:		Date:	

What went well this week (be specific — cite an example):

What to improve, with one concrete suggestion:

Recommended focus for next week:

Any concerns needing program-lead attention? ☐ No ☐ Yes — notes below

9.10 KPI Rating Scale & Grade Conversion

Score	Rating	Description
5	Outstanding	Consistently exceeds expectations; work is close to industry-ready quality; proactively helps others.
4	Exceeds Expectations	Regularly goes beyond the minimum; strong independent problem-solving.
3	Meets Expectations	Solid, reliable performance at the level expected for an intern at this stage.
2	Below Expectations	Inconsistent quality or engagement; needs closer support to hit weekly goals.
1	Needs Significant Improvement	Struggling to meet basic requirements; needs an active improvement plan.

Overall weighted percentage (from Section 9.7) maps to a final program rating as follows:

Overall Score	Final Rating	Typical Outcome
90–100%	Outstanding	Certificate + strong reference + priority consideration for future roles
80–89%	Exceeds Expectations	Certificate + reference
70–79%	Meets Expectations	Certificate issued
55–69%	Needs Improvement	Certificate of Participation only; improvement plan discussed
Below 55%	Unsatisfactory	No certificate; program-lead review

10. Appendices

Appendix A — Consolidated Master Resource List

Every resource cited throughout this roadmap, grouped by category for quick reference. Track-specific context for each link is in its topic cluster in Sections 2 and 4–8.

Official Documentation

Resource	URL
Python	https://docs.python.org/3/
Git	https://git-scm.com/doc
GitHub Docs	https://docs.github.com/en/get-started
Docker	https://docs.docker.com/
Ubuntu	https://help.ubuntu.com/
Cisco Networking Academy	https://www.netacad.com/
Wireshark	https://www.wireshark.org/docs/
NIST Cybersecurity Framework	https://www.nist.gov/cyberframework
NIST CSRC	https://csrc.nist.gov/
MITRE ATT&CK	https://attack.mitre.org/matrices/enterprise/
OWASP Top 10	https://owasp.org/Top10/
Wazuh	https://documentation.wazuh.com/current/index.html
Elastic (ELK Stack)	https://www.elastic.co/guide/index.html
Splunk	https://docs.splunk.com/Documentation/Splunk
SigmaHQ	https://github.com/SigmaHQ/sigma
OpenStack	https://docs.openstack.org/
Kolla Ansible	https://docs.openstack.org/kolla-ansible/latest/
DevStack	https://docs.openstack.org/devstack/latest/
Zabbix	https://www.zabbix.com/documentation/
Grafana	https://grafana.com/docs/
LibreNMS	https://docs.librenms.org/
Prometheus	https://prometheus.io/docs/
HashiCorp Terraform	https://developer.hashicorp.com/terraform
Ansible	https://docs.ansible.com/
Nmap Reference Guide	https://nmap.org/book/man.html
BloodHound	https://bloodhound.specterops.io/

Resource	URL
Metasploit	https://docs.metasploit.com/
GTFOBins	https://gtfobins.github.io/
John the Ripper	https://www.openwall.com/john/
NetExec (formerly CrackMapExec)	https://github.com/Pennyw0rth/NetExec
Android Developers	https://developer.android.com/
Kotlin	https://kotlinlang.org/docs/home.html
Firebase	https://firebase.google.com/docs
Apple Developer Documentation	https://developer.apple.com/documentation/
OpenAI API	https://platform.openai.com/docs
LangChain	https://python.langchain.com/
Hugging Face	https://huggingface.co/
Kubernetes	https://kubernetes.io/docs/home/
Trivy	https://github.com/aquasecurity/trivy
Selenium	https://www.selenium.dev/documentation/
Playwright	https://playwright.dev/
PowerShell (Microsoft Learn)	https://learn.microsoft.com/en-us/powershell/
Sysmon (Sysinternals)	https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon
Markdown Guide	https://www.markdownguide.org/

Free Courses & Learning Platforms

Resource	URL
Linux Foundation — Introduction to Linux (LFS101)	https://training.linuxfoundation.org/training/introduction-to-linux/
freeCodeCamp	https://www.freecodecamp.org/
Cisco Skills for All	https://www.netacad.com/
PortSwigger Web Security Academy	https://portswigger.net/web-security
roadmap.sh	https://roadmap.sh/
roadmap.sh — DevSecOps	https://roadmap.sh/devsecops
roadmap.sh — AI Engineer	https://roadmap.sh/ai-engineer
roadmap.sh — Android	https://roadmap.sh/android
Atlassian Agile Coach	https://www.atlassian.com/agile
GitHub Skills	https://skills.github.com/
Real Python	https://realpython.com/

Hands-On Practice Platforms

Resource	URL
OverTheWire — Bandit wargame	https://overthewire.org/wargames/bandit/
TryHackMe	https://tryhackme.com/
HackTheBox Academy	https://academy.hackthebox.com/catalogue/paths
LetsDefend	https://letsdefend.io/
MITRE ATT&CK Navigator	https://mitre-attack.github.io/attack-navigator/
Play with Docker	https://labs.play-with-docker.com/
OSINT Framework	https://osintframework.com/
Wireshark Sample Captures	https://wiki.wireshark.org/SampleCaptures
VirusTotal	https://www.virustotal.com/

Repositories

Resource	URL
first-contributions (beginner PR practice)	https://github.com/firstcontributions/first-contributions
openstack/kolla-ansible	https://github.com/openstack/kolla-ansible
BloodHound	https://github.com/SpecterOps/BloodHound
THC-Hydra	https://github.com/vanhauser-thc/thc-hydra

Resource	URL
NetExec	https://github.com/Pennyw0rth/NetExec
Trivy	https://github.com/aquasecurity/trivy

Certifications (Optional)

Certification	URL
CompTIA (Linux+, Network+, Security+)	https://www.comptia.org/
Cisco CCNA	https://www.cisco.com/site/us/en/learn/training-certifications/certifications/enterprise/ccna/index.html

Certifications are optional stretch goals, not program requirements. Interns interested in pursuing one should discuss timing with their mentor — most are best attempted after, not during, the internship.

Appendix B — Required Tools & Software Checklist

For lab-environment setup ahead of Day 1. All tools are free / open-source or have a free tier sufficient for this program.

Phase 1 — All Interns

Tool	Used For
VirtualBox or VMware Workstation Player	Local VM hosting for Linux/Docker labs
A Linux distribution ISO (e.g., Ubuntu Server/Desktop)	Primary Linux lab environment
Wireshark	Packet capture and analysis
Cisco Packet Tracer	Basic networking topology labs
Python 3.x + pip	Programming and automation labs
Git + a GitHub account	Version control for every deliverable
Docker Desktop / Docker Engine	Containerization labs
A code editor (e.g., VS Code)	Writing scripts, configs, and documentation

Track 1 — NOC

- Cisco Packet Tracer
- GNS3 (+ a router/switch image set)
- Wireshark
- Zabbix
- Grafana
- LibreNMS

Track 2 — SOC

- Wazuh
- ELK Stack (Elasticsearch, Logstash, Kibana)
- Splunk (free tier)
- Microsoft Sentinel (trial/lab tenant)
- Velociraptor
- Sysmon
- VirusTotal account

Track 3 — Cloud (OpenStack)

- DevStack-compatible Linux VM (min. 8GB RAM recommended)
- Kolla-Ansible
- OpenStack CLI (python-openstackclient)
- Terraform
- Ansible
- Prometheus + Grafana

Track 4 — Red Team

- Kali Linux (VM or bare metal)
- Burp Suite Community Edition
- Metasploit Framework
- BloodHound + Neo4j
- NetExec
- Gobuster
- Hydra
- John the Ripper

Track 5 — R&D

- Domain-dependent — see below

R&D domain-specific tools: AI/LLMs — Python, LangChain, Hugging Face account, OpenAI-compatible API key.

Android — Android Studio, Kotlin. iOS — Xcode (macOS required), Swift. DevSecOps — GitHub Actions or Jenkins, Docker, Kubernetes (e.g., kind/minikube), Trivy, SonarQube (Community Edition). Automation — Selenium or Playwright, PowerShell.

Appendix C — Glossary of Acronyms

Term	Meaning
ACL	Access Control List — rules that permit or deny traffic
AD	Active Directory — Microsoft's directory service for Windows networks
API	Application Programming Interface
ATT&CK	Adversarial Tactics, Techniques & Common Knowledge (MITRE framework)
CIA Triad	Confidentiality, Integrity, Availability — core security model
CI/CD	Continuous Integration / Continuous Deployment
CLI	Command-Line Interface
CVSS	Common Vulnerability Scoring System
DHCP	Dynamic Host Configuration Protocol
DNS	Domain Name System
EDR	Endpoint Detection & Response
ELK	Elasticsearch, Logstash, Kibana (log analytics stack)
HA	High Availability
HOT	Heat Orchestration Template (OpenStack)
IaC	Infrastructure as Code
IaaS	Infrastructure as a Service
IOC	Indicator of Compromise
IR	Incident Response
IPsec	Internet Protocol Security (VPN encryption standard)
LLM	Large Language Model
MTTD / MTTR	Mean Time to Detect / Mean Time to Respond (or Repair)
MVVM	Model-View-ViewModel (software architecture pattern)
NAT / PAT	Network Address Translation / Port Address Translation
NOC	Network Operations Center
NSE	Nmap Scripting Engine
OSI Model	Open Systems Interconnection model (7-layer networking reference)
OSINT	Open-Source Intelligence
OSPF	Open Shortest Path First (routing protocol)
PKI	Public Key Infrastructure
PTES	Penetration Testing Execution Standard
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer (API architecture style)

Term	Meaning
SIEM	Security Information & Event Management
SNMP	Simple Network Management Protocol
SOAR	Security Orchestration, Automation & Response
SOC	Security Operations Center
SSH	Secure Shell
STP	Spanning Tree Protocol
TCP/IP	Transmission Control Protocol / Internet Protocol
VLAN	Virtual Local Area Network
VPN	Virtual Private Network